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A Flexible Composite Term Structure of Inflation Expectations

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A Flexible Composite Term Structure of Inflation Expectations

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ABSTRACT: We develop a flexible methodology for estimating a real-time term structure of inflation expectations from heterogeneous survey- and market-based sources. Our Nelson-Siegel Internally Consistent Expectations model extends the standard dynamic Nelson-Siegel state-space model by imposing a restriction that forecasts of any fixed-horizon event are revised only in response to new information, and by calibrating measurement-error variances to each source's historical forecast accuracy. The resulting composite summarizes expectations at all horizons in a single internally-consistent term structure that can be applied in a variety of contexts, including when data sources are reported at irregular intervals and horizons. To illustrate our framework, we apply the model to Japanese inflation expectations data, and document a gradual alignment of long-run inflation expectations with the Bank of Japan's 2-percent target from below.

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WORKING PAPERS

A Flexible Composite Term Structure of Inflation Expectations

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1 Introduction

Inflation expectations are a central determinant of price setting decisions in standard New Keynesian models (Woodford, 2003; Galí, 2015) and are leading empirical predictors of future inflation (Faust and Wright, 2013). Their dynamics are crucial for a range of macroeconomic assessments, including the persistence of inflation, the degree of central bank credibility, and monetary policy transmission (Bems and others, 2021; Carrière-Swallow and others, 2021). Central bankers therefore pay considerable attention to monitoring them, and they feature prominently in policy communications.

Monitoring inflation expectations in real time is complicated by the fact that they are measured indirectly by a range of surveys and market-based indicators—covering the expectations of households, firms, professional forecasters, and financial markets—that differ in their reporting horizons, sampling frequency, and predictive accuracy. This heterogeneity can hamper inference about aggregate expectations, particularly when the question of interest relies on expectations across multiple horizons, which no single source may measure directly. Estimating a composite term structure of expectations from these indicators is, therefore, an important measurement problem for monetary policymakers.

This paper develops a flexible methodology to estimate the term structure of inflation expectations by combining information across these heterogeneous sources. We propose the Nelson-Siegel Internally Consistent Expectations (NSICX) model, an extension of the standard dynamic Nelson-Siegel (DNS) state-space model in the spirit of Nelson and Siegel (1987) and Diebold and Li (2006). The DNS framework has previously been applied to inflation expectations by Aruoba (2020), whose Aruoba Term Structure of Inflation Expectations (ATSIX) model is estimated on US professional forecasts from Blue Chip Economic Indicators and the Survey of Professional Forecasters. Our implementation departs from ATSIX in two main respects: we impose an internal-consistency restriction, under which the factor dynamics are pinned down by the cross-sectional loading structure so that forecasts of any fixed-horizon event are revised only in response to new information,[1] and by using a weighting scheme that combines sources in proportion to their historical forecast accuracy. The model's output is a panel of inflation expectations indexed by a cross-sectional dimension spanning horizons from 3 to 120 months, and a time-series dimension observed at a monthly frequency.

First, we specify the transition equation to be internally consistent with the Nelson-Siegel (NS) loading structure, deriving the unique transition dynamics under which forecast revisions for any fixed inflation event are zero in the absence of new shocks. The resulting dynamics imply a non-stationary level factor that follows a random walk, while the slope and curvature factors decay at rate λ, which also governs how curvature feeds into the slope. Because both the transition dynamics and the NS loadings depend on λ, identification of this parameter draws on both the time-series and cross-sectional properties of the latent factors. This is valuable because the cross-section alone often identifies λ weakly, in that a wide range of empirically relevant values fit the observed term

[1] A related consistency requirement arises in arbitrage-free NS models for bond yields (Christensen and others, 2011), though there it is imposed under the risk-neutral measure, in contrast to the physical measure relevant for internally-consistent expectations.

structure almost equally well (as we document for Japan). The ATSIX model, by contrast, specifies independent stationary AR(3) processes for each factor. Whether this restriction is appropriate is ultimately an empirical question: Section 4.4 documents prolonged shifts in trend inflation in Japan, consistent with a nominal anchor that moves over time as the economy or policy regime changes, and better captured with a unit-root level factor. A formal variance ratio test supports this distinction, rejecting the unit-root null in favor of stationarity for the US level factor but not for Japan's.

Second, reconciling information across multiple sources and horizons requires a principled weighting scheme, particularly when these sources display heterogeneous biases or are reported irregularly. For each source i and horizon h, we calibrate measurement error variances $\sigma_i^2(h)$ to be proportional to the mean squared forecast error (MSFE) of source i at horizon h relative to a statistical benchmark. Normalizing MSFEs in this way ensures comparability across sources and horizons with unequal evaluation samples, which can arise from differences in data availability and from the fact that long-horizon forecasts can only be evaluated on older realizations. The objective of this weighting approach is to extract the common signal across heterogeneous sources in a simple and transparent way, while prioritizing forecasting performance.[2]

An important consequence of these modeling choices is that the state-space parameters reduce to four, $\theta = (q_L, q_S, q_C, \lambda)$, jointly estimated by maximum likelihood, compared to 16 in ATSIX (excluding measurement-error variances). Parsimony matters in state-space models because the factors are latent, so more flexible transition parametrizations can overfit in sample by attributing spurious variation to latent dynamics, yielding an inaccurate summary of underlying expectations. The Aruoba (2020) application is more favorable on the measurement side, as they estimate their model on only two sources of inflation expectations, but more generally, the case for parsimony is stronger when the measurement of expectations involves multiple sources reported at irregular intervals and horizons, a structure common to many economies.

Japan is a setting with exactly this structure: multiple sources reported at irregular intervals and horizons. We combine six sources of inflation expectations into a single aggregate composite, and these sources differ markedly in levels.[3] Household expectations from the Bank of Japan (BOJ) Opinion Survey have historically been above target and have increased further in recent years. Firm expectations from the BOJ Tankan Survey have also moved above 2 percent. Professional forecasters' medium- and long-term expectations remain somewhat below target, while market-implied 10-year breakeven inflation (BEI) rates have moved to about 2 percent.[4]

This pattern, whereby household and firm expectations exceed those of professional forecasters and financial markets, is consistent with stylized facts for the US (Weber and others, 2022), and

[2]Alternative objectives are possible, including weighing sources according to their relevance for aggregate price-setting behavior. We leave this exploration for future research.

[3]We do not incorporate qualitative or directional measures, nor higher-order moments of the expectation distribution such as dispersion, even though these measures have been shown to provide additional predictive information for inflation in the United States (Mankiw and others, 2004; Brandão-Marques and others, 2023; Fofana and others, 2024; Binder, 2017).

[4]The BOJ formally adopted the 2-percent inflation target in January 2013. Comparisons of expectations to the target prior to that date are made for reference and should be interpreted with this caveat in mind.

similar heterogeneity across data sources has also been documented for Japan by Maruyama and Suganuma (2020). Despite this heterogeneity in levels, we find that expectations share substantial common variation across sources and horizons. After standardization, the first principal component across sources and horizons accounts for 91 percent of the total variation with near-uniform loading across all source-horizon pairs, indicating a common level factor. The second brings the cumulative share to over 96 percent and loads monotonically across horizons, capturing term-structure variation; the third reflects small residual cross-source disagreement. We show that a three-factor NS loading structure fits the cross-section as well as the two-component unrestricted PCA, with slope and curvature jointly captured by the second component, reflecting their co-movement at a quarterly frequency. These results support a common latent term structure for inflation expectations across sources, well captured by the low-dimensional NS representation.

For most of the preceding two decades, the estimated term structure for Japan remained close to flat at near-zero levels, but since 2022, our estimates indicate that inflation expectations across all horizons have increased meaningfully. Medium-term expectations at the two-year horizon, which correspond roughly to the BOJ's policy horizon, have risen steadily, with the two-percent target entering the estimated 95-percent confidence band by mid-2022. Long-term expectations show a similar upward trend, entering the band by late 2024. While uncertainty remains, reflecting model specification choices, data input sensitivity, and limited historical variation in expectations following a prolonged period of near-zero inflation, the evidence is consistent with inflation expectations gradually aligning with the BOJ's 2-percent target from below.

We next compare our real-time trend inflation expectation for Japan against its US counterpart, the regularly updated ATSIX estimates (Federal Reserve Bank of Philadelphia, 2026). The two series display a striking divergence over the sample. US trend inflation remained close to 2 percent throughout the post-GFC period; Japan, by contrast, remained persistently below 2 percent for most of the sample, rising toward the 2-percent target only near the end. In particular, we find that monthly innovations to trend expectations are mean-reverting in the US, but largely permanent in Japan, consistent with the prolonged shifts in the inflation regime documented for Japan.

To formally examine the anchoring of long-term expectations, we regress quarterly innovations in trend expectations on proxies for short-run inflation surprises. Japan's long-run expectations respond significantly to short-run inflation surprises and near-term expectation revisions. While we do not attempt to explain the drivers of expectations in this paper, this pattern is consistent with long-term expectations in Japan on average responding to realized inflation outcomes, alongside revisions in near-term expectations. The US also shows a positive and significant response to near-term revisions, but with a smaller sensitivity coefficient than the corresponding estimate for Japan. The full-sample comparison, however, masks substantial time variation: rolling-window estimates show Japan's coefficient was close to 0.5 in the early 2000s, implying a passthrough of close to 50 percent from short-run to long-run inflation expectations. This has declined to about 0.08 by 2026, broadly in line with the contemporaneous US estimate of 0.04.

Additionally, we use the composite term structure to construct ex-ante real interest rates across maturities. Ex-ante real rates, computed as nominal Japanese Government Bonds (JGB) yields

minus our composite inflation expectations, differ from inflation-indexed JGB (JGBi) real yields, with the wedge varying over time in a manner broadly consistent with estimates of premia on JGBi bonds from Christensen and Spiegel (2024). The time series of real rates reveals a persistent shift into negative territory following the introduction of quantitative and qualitative easing in April 2013, with short-term real rates reaching approximately -2.3 percent in 2014, and about -2.7 percent during the post-pandemic episode in early 2024, reflecting increases in inflation expectations ahead of the 2024 hiking cycle.

Our paper connects to several strands of the literature. Most directly, it is related to the estimation of expectation term structures, which includes the ATSIX model of Aruoba (2020). Chernov and Mueller (2012) also estimate an expectations term structure that leverages survey forecasts, but they do so within a bond-pricing model in which identification rests primarily on the pricing of risk factors, including inflation. Our work also relates to Giacomini and others (2025), who study the term structure of expectation *revisions* (as opposed to levels, which reflect accumulated revisions). They estimate perceived impulse responses from fixed-event forecast revisions, and their perceived shocks are closely related to the internally-consistent NS factor innovations in our framework. Notably, they highlight that the ATSIX transition model fails to satisfy the same notion of internal consistency that we impose, which prevents it from properly fitting observed fixed-event revisions in sample.

Another related strand evaluates and combines alternative measures of expectations. Faust and Wright (2013) find professional consensus forecasts difficult to outperform in inflation forecasting. Relatedly, Ang and others (2007) show that consensus survey forecasts outperform a wide range of alternatives and that accuracy-weighted combinations perform best out-of-sample, an approach our relative-MSFE weighting extends to the term structure.

For our analysis of expectation anchoring, we note that there is no single agreed-upon criterion in the literature. We assess it for Japan under two complementary notions: whether the estimated long-run expectation aligns with the central bank's target and whether it is sensitive to short-run inflation developments.[5] However, we acknowledge that broader and more demanding definitions could lead to different conclusions about anchoring, including for example Coibion and Gorodnichenko (2026)'s surprising finding of weakly anchored expectations in the United States.

We also contribute to research on Japanese inflation expectations (Maruyama and Suganuma, 2020; Osada and Nakazawa, 2024; Christensen and Spiegel, 2024). The latent-factor model of Christensen and Spiegel (2024) estimated that Japanese long-run expectations sat below the BOJ's 2-percent target, whereas our survey-dependent composite estimate points to a gradual alignment toward target from below over the past few years. Our analysis of ex-ante real rates also relates to previous work studying the role of inflation risk, liquidity premium, and deflation protection in nominal long-term bond yields (D'Amico and others, 2018; Haubrich and others, 2012).

The remainder of the paper proceeds as follows. Section 2 presents the NSICX state-space

[5] We proxy for these developments using inflation surprises, which is common in the literature (Gürkaynak and others, 2010; Beechey and others, 2011; Bems and others, 2021; Mehrotra and Yetman, 2018; Crump and others, 2023; Giacomini and others, 2025).

term structure model. Section 3 describes the data sources and evaluates their comovement and forecasting performance. Section 4 reports the estimated term structure and presents applications to trend inflation comparisons between Japan and the US and to model-implied real interest rates. Section 5 concludes.

2 Model

Our NSICX model is a special case of a general linear Gaussian state-space system, in which a small number of latent factors govern the term structure of inflation expectations across all sources and horizons simultaneously (Hamilton, 1994; Harvey, 1989; Durbin and Koopman, 2012). The measurement equation links observed inflation expectations to the latent factors through Nelson-Siegel loadings, while the transition equation governs their time-series evolution. Conditional on the model parameters, the system is operationalized via the Kalman filter, which produces filtered (real-time) and smoothed (full-sample) estimates of the latent factors by optimally weighting all available observations at each date. This section describes the NS term structure, the measurement and transition equations, and the estimation procedure in turn.

2.1 Nelson-Siegel Term Structure

We define the expected instantaneous forward inflation rate at horizon h as

$$f_t(h) = -\frac{\partial}{\partial h} \log P_t(h), \tag{1}$$

where $P_t(h)$ is the expected price level at $t + h$.

Let $\pi_t(h)$ denote inflation expectations at date t over horizon h months, defined using continuous compounding as the annualized log change in the price level from month t to month $t + h$. This is given by the integral of the forward rate:

$$\pi_t(h) = \frac{1}{h} \int_0^h f_t(s)\, ds. \tag{2}$$

We model the instantaneous forward rate using the NS functional form,

$$f_t(h) = L_t + e^{-\lambda h} S_t + \lambda h e^{-\lambda h} C_t, \tag{3}$$

so that integrating (3) according to (2) yields the familiar NS form for average inflation expectation

$$\pi_t(h) = L_t - \frac{1 - e^{-\lambda h}}{\lambda h} S_t + \left(\frac{1 - e^{-\lambda h}}{\lambda h} - e^{-\lambda h} \right) C_t, \tag{4}$$

where L_t, S_t, and C_t are the level, slope, and curvature factors, and $\lambda > 0$ is a shape parameter. L_t captures long-run inflation expectations, S_t captures the spread between long- and short-run expectations, and C_t captures medium-term deviations relative to the short and long ends of the term

structure. Note that as $h \to \infty$, the forward rate converges to L_t, consistent with its interpretation as the expected trend inflation rate.

We find the choice of functional form for inflation expectations is appropriate for the following reasons. First, NS provides a parsimonious three-factor representation with a transparent level, slope, and curvature interpretation and bounded long-run forward rates. Second, this low-dimensional representation is consistent with the data: in 3.3.1, we find that three NS factors fit the cross-sectional variation in observed expectations as well as an unconstrained three-factor PCA.

A key advantage of the continuous-compounding definition of inflation is that it preserves the linearity of the state-space system. In particular, expected inflation over any forward window $[h_1, h_2]$ with $h_2 > h_1 \geq 0$ is the average of the instantaneous forward rates over that window,

$$\pi_{t+h_1 \to t+h_2|t} = \frac{1}{h_2 - h_1} \int_{h_1}^{h_2} f_t(s)\, ds, \tag{5}$$

which, upon substituting (3) and integrating, yields the closed-form expression

$$\pi_{t+h_1 \to t+h_2|t} = L_t + \frac{e^{-\lambda h_1} - e^{-\lambda h_2}}{\lambda(h_2 - h_1)}(C_t - S_t) + \frac{h_1 e^{-\lambda h_1} - h_2 e^{-\lambda h_2}}{h_2 - h_1} C_t. \tag{6}$$

The special case $h_1 = 0$ corresponds to (4), and the limit $h_1 \to h_2$ recovers the instantaneous forward rate at $h = h_1 = h_2$ in (3).

2.2 Measurement Equations

Each observed inflation expectation measure is treated as a noisy signal of the underlying term structure at the relevant horizon. Let $y^i_{t,h}$ denote the observation from source i at date t reported at horizon h. The measurement equation for a single observation is

$$y^i_{t,h} = x(h;\lambda)'\beta_t + \varepsilon^i_{t,h}, \qquad \varepsilon^i_{t,h} \sim N\big(0,\, \sigma_i^2(h)\big)\,,$$

where $\beta_t = (L_t, S_t, C_t)'$ is the vector of latent factors and $x(h;\lambda)$ are the corresponding NS factor loadings from (4). For sources that report expectations over a forward window $[h_1, h_2]$, the loadings are replaced by the forward-average loadings implied by (5).

Stacking all n_t observations available at date t gives the system-level measurement equation,

$$y_t = Z_t(\lambda)\beta_t + \varepsilon_t, \qquad \varepsilon_t \sim N(0,\, R_t), \quad R_t = \mathrm{diag}\{\sigma_i^2(h)\}, \tag{7}$$

where $Z_t(\lambda)$ is the $n_t \times 3$ matrix of NS loadings whose i-th row is $x(h_i;\lambda)'$ for a point-horizon observation or the forward-average loadings from (5) for a window observation. The matrix $Z_t(\lambda)$ is time-varying because the set of available sources and horizons n_t changes across periods, accommodating the unbalanced panel structure of the data, and (7) enters the system directly via the Kalman filter.

2.3 Transition Equations

The transition equation is a key departure from the ATSIX model. Internal consistency of the DNS framework requires factor dynamics to align with the cross-horizon restrictions implied by the Nelson-Siegel loading structure. To derive the transition equation that satisfies internal consistency, consider a fixed event defined over the window $[t+h_1, t+h_2]$. The expectation for this event, formed at time t, is $\pi_{t+h_1 \to t+h_2|t}$, which is a linear function of the time t NS factors via (6). One period later, the expectation for the same fixed event, now formed at $t+1$, corresponds to the forward window $[h_1 - 1, h_2 - 1]$. Using (6), the revision to this fixed-event expectation is

$$\begin{aligned}
&\pi_{t+h_1 \to t+h_2|t+1} - \pi_{t+h_1 \to t+h_2|t} \\
&\quad = \Delta L_t + \frac{e^{-\lambda(h_1-1)} - e^{-\lambda(h_2-1)}}{\lambda(h_2 - h_1)}(C_{t+1} - S_{t+1}) + \frac{(h_1-1)e^{-\lambda(h_1-1)} - (h_2-1)e^{-\lambda(h_2-1)}}{h_2 - h_1} C_{t+1} \\
&\quad\quad - \frac{e^{-\lambda h_1} - e^{-\lambda h_2}}{\lambda(h_2 - h_1)}(C_t - S_t) - \frac{h_1 e^{-\lambda h_1} - h_2 e^{-\lambda h_2}}{h_2 - h_1} C_t. \qquad (8)
\end{aligned}$$

Internal consistency requires that (8) equals zero for all choices of $[t + h_1, t + h_2]$ absent innovations, i.e., when $\eta_t = 0$ so that $\beta_{t+1} = F\beta_t$. Substituting $F\beta_t$ for β_{t+1} in (8) and setting the revision to equal zero yields a system of constraints on F. The unique solution is

$$F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & a & \lambda a \\ 0 & 0 & a \end{pmatrix}, \qquad a = e^{-\lambda}, \qquad (9)$$

which can be verified by direct substitution: with $L_{t+1} = L_t$, $S_{t+1} = aS_t + \lambda a C_t$, and $C_{t+1} = aC_t$, the forward loadings at horizon $[h_1 - 1, h_2 - 1]$ evaluated at β_{t+1} reproduce exactly the loadings at $[h_1, h_2]$ evaluated at β_t, so (8) equals zero.

We therefore model the NS factors to evolve according to

$$\beta_t = F\beta_{t-1} + \eta_t, \qquad \eta_t \sim N(0, Q), \qquad (10)$$

where $Q = \mathrm{diag}(q_L, q_S, q_C)$ and F is given by (9). The level factor follows a random walk, consistent with its role as the sole permanent component. The slope and curvature factors mean-revert at rate $a = e^{-\lambda}$, with off-diagonal coupling λa uniquely pinned down by the NS structure. The transition matrix F is fully determined by λ, which is already required for the measurement equation, and therefore introduces no additional free parameters.

2.4 Estimation

The measurement error variances $\sigma_i^2(h)$ are calibrated externally from historical forecast performance rather than estimated jointly with the other model parameters. The calibrated values for Japan and their implied weights are reported in Section 3.

The remaining model parameters are estimated by maximum likelihood via the prediction-error

decomposition of the Kalman filter.[6] The free parameters are $\theta = (q_L, q_S, q_C, \lambda)$, all constrained to be positive, with measurement error variances held fixed at their calibrated values. The filter is initialized using the exact diffuse initialization of Durbin and Koopman (2012), and the first 12 months of the sample are excluded from reported estimates as a precaution against residual effects of the diffuse initialization, which may be amplified by the relatively sparse observation coverage at the start of the sample. To guard against local optima, which can arise from the multimodal likelihood surface induced by λ, the optimization is run from multiple starting values using the L-BFGS-B algorithm. Filtered factor estimates condition only on data available at each date and thus reflect the real-time availability of inflation expectations, while smoothed estimates are obtained by running the Kalman smoother over the full sample and incorporate ex-post information. Pointwise confidence bands are constructed using the Durbin-Koopman simulation smoother. We draw $B = 500$ parameter vectors from $N(\hat{\theta}, \hat{V})$, where $\hat{\theta}$ is the MLE and $\hat{V}$ is the asymptotic covariance matrix estimated via the outer product of gradients (OPG), truncating draws that violate the positivity constraints. For each draw, we re-run the Kalman smoother and report empirical 68- and 95-percent bands across draws.[7]

3 Data

3.1 Expectation Sources

We draw on six sources of inflation expectations covering households, firms, professional forecasters, and financial markets, spanning horizons from 3 to 120 months. The sample varies by indicator: household expectations start in 2004, firm surveys start in 2016, professional forecaster surveys are used from 2004 onward, and financial market data starts in 2004.[8] The number of observations per period was more limited at the start of the sample but has been broadly stable since around 2016. Figure 1 summarizes the frequency of observations by source and horizon. The majority of measurements have horizons below 24 months, with additional clusters at 5- and 10-year horizons; only Consensus Economics and BEI expectations at the 10-year horizon.[9]

[6] Durbin and Koopman (2012) provides a textbook treatment of the prediction-error decomposition, the exact diffuse initialization, and the simulation smoother. The model is implemented as a subclass of the `MLEModel` state space framework in `statsmodels` (Fulton, 2017), which provides Kalman filtering and MLE via numerical optimization once the system matrices are specified.

[7] The filtered state estimates reflect real-time inflation expectations conditional on model parameters and calibrated measurement error variances, both of which are estimated over the full sample, introducing a degree of look-ahead at the parameter stage. Real-time vintages of the estimates could in principle be constructed by re-estimating parameters and calibrations on an expanding window, but we leave this for future work.

[8] Consensus Economics monthly forecasts for Japan are available from 1989, but we begin our sample in 2004, when the other principal sources (BOJ Opinion Survey, JGBi BEI) become available.

[9] The selection of sources broadly aligns with Osada and Nakazawa (2024), with three differences. We use BEI rates rather than inflation swap rates, given that the latter is more thinly traded in Japan. The QUICK survey is omitted due to a lack of data access. We further exclude qualitative household expectations from the BOJ Opinion Survey, as their conversion into a quantitative series may be sensitive to the parametric imputation assumptions.

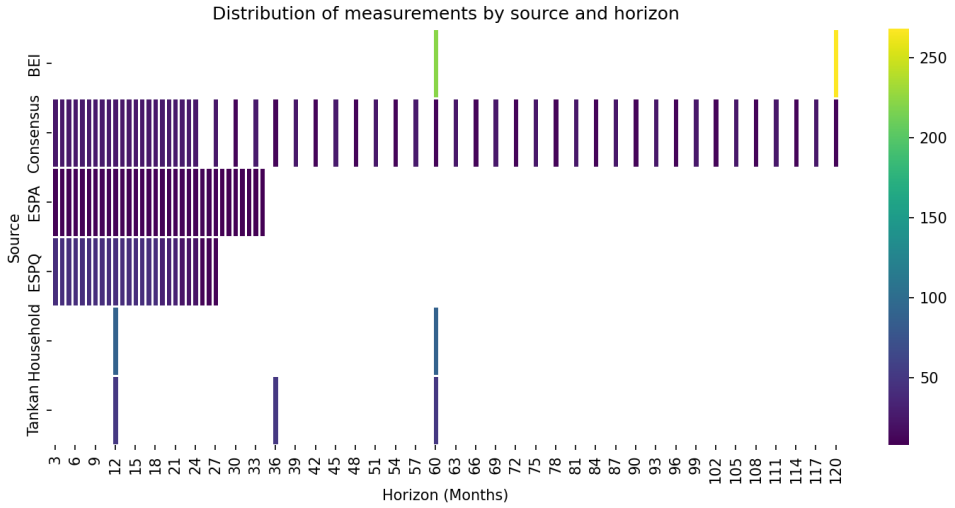


Figure 1: Data availability by source and horizon. Each row corresponds to a source of inflation expectations; shaded regions indicate horizons in which at least one observation was available at some point in time.

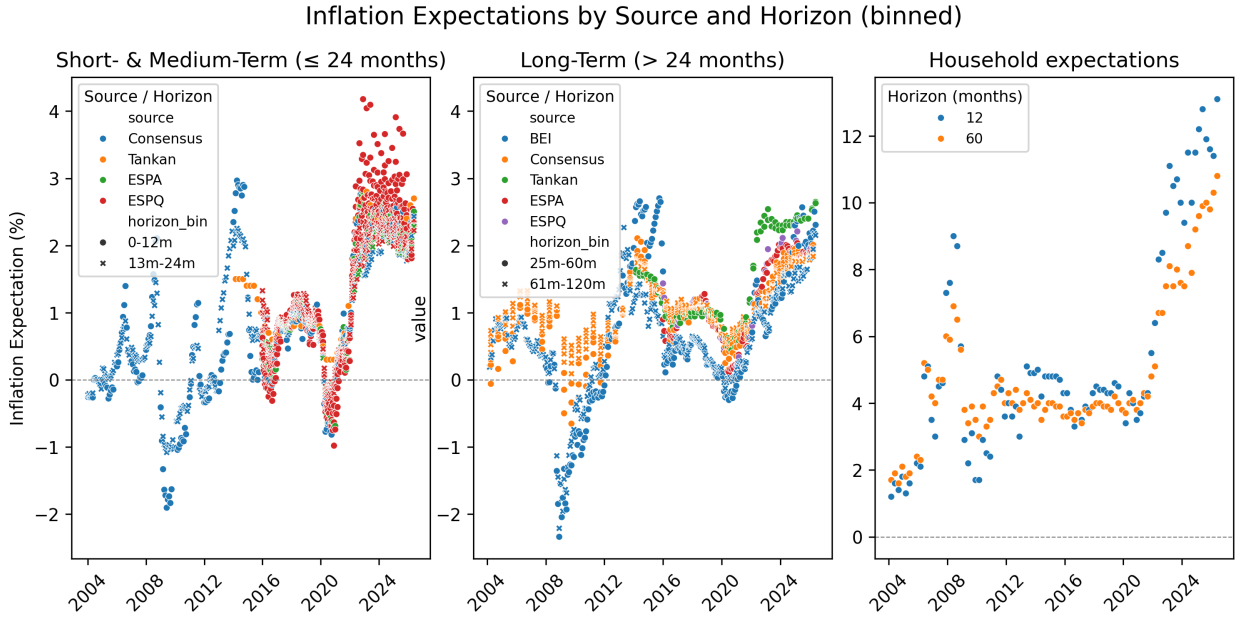


Figure 2: Inflation expectations by source and horizon. All expectations are expressed as annualized percentage rates.

3.1.1 Household expectations

Household inflation expectations are drawn from the Bank of Japan's Opinion Survey on the General Public's Views and Behaviors. The survey reports quantitative expectations for inflation over the next year and the average expected inflation over the next five years (annualized). The indicator is available at the quarterly frequency starting from 2004Q1.

3.1.2 Firm expectations

Firm inflation expectations are drawn from the Bank of Japan's Tankan Survey, which surveys firms across a range of industries. The survey reports expectations for general price increases over the next year, three years ahead (annualized), and five years ahead (annualized). A notable feature of the Tankan Survey format is that the reported forecasts correspond to average expected inflation over a fixed 12-month window ending at the stated horizon, rather than cumulative average inflation from the current date. Accordingly, Tankan observations at the 1-, 3-, and 5-year horizons are treated in the model as spanning the window $[h-12, h]$ for $(h = 12, 36, 60)$. The measure is available at the quarterly frequency, starting from 2016Q1.

3.1.3 Professional forecaster expectations

Professional forecaster expectations are drawn from two sources. Consensus Economics reports analyst expectations for CPI headline inflation up to 10-years ahead. Expectations up to 2-years ahead are available monthly back to 1989, but we use the sample starting from January 2004 to align with the start of our other sources; expectations from 3- to 10-years ahead are available every quarter starting from 2012. Meanwhile, the ESP Forecast Survey published by the Japan Center for Economic Research (JCER) is available at both annual and quarterly frequencies (ESPA and ESPQ respectively). These survey data are available starting from 2016. For both Consensus and ESP, we use the cross-sectional mean of individual forecaster responses as the input to the model.

3.1.4 Market-implied expectations

Market-implied inflation expectations consist of breakeven inflation (BEI) rates, computed as the spread between nominal Japanese Government Bonds (JGBs) and inflation-indexed JGBs at 5- and 10-year maturities. These data are available at the daily frequency, starting from 2004 for the 10-year horizon and 2008 for the 5-year horizon, and we aggregate them into monthly averages for use in the model.

3.1.5 Statistical benchmark

To provide a common yardstick for evaluating forecast accuracy across sources and horizons, we compare each source's MSFE against that of a simple rolling-mean forecast. For each month t, the benchmark forecast is the trailing 12-month average of year-on-year headline CPI inflation observed up to month $t-1$. This choice accommodates the typical publication delay for CPI;

therefore, by construction, the benchmark is backward-looking and uses only information available at the forecast origin date, thereby avoiding look-ahead bias.

$$\hat{\pi}_{t+h|t} = \frac{1}{12}\sum_{j=1}^{12} \pi^{\text{yoy}}_{t-j}, \qquad \pi^{\text{yoy}}_{t-j} = \left(\ln P_{t-j} - \ln P_{t-j-12}\right) \times 100, \tag{11}$$

where P_t is the headline CPI index, and the same forecast is used for all horizons $h = 1, \ldots, 132$ months.

3.2 Data Transformations and Forecast Performance

All inflation expectations are converted (when necessary) to be expressed as forward-looking annualized percentage rates. In particular, both Consensus and ESP expectations are reported as fixed-event calendar-year forecasts rather than fixed-horizon forecasts. These are converted to fixed-horizon forward-looking expectations by stripping out the portion of the calendar year that has already been realized. Specifically, for a forecast made in month t for a target year Y, with k months already realized and $J = 12 - k$ months remaining, the implied J-month forward-looking expectation is

$$m^{\text{fwd}}_{i,t}(J) \;=\; \frac{12}{J}\, F_{i,t}(Y) \;-\; \frac{1}{J}\sum_{m=1}^{k} \pi_m, \tag{12}$$

where $F_{i,t}(Y)$ is the reported fixed-event forecast and π_m are realized monthly inflation rates within the target year. When $k = 0$, the forward-looking expectation reduces to the fixed-event forecast itself: $m^{\text{fwd}}_{i,t}(12) = F_{i,t}(Y)$.

For forecasts made more than 12 months before the end of the target year ($J > 12$), the forward-looking expectation is recovered recursively:

$$m^{\text{fwd}}_{i,t}(J) \;=\; \frac{12}{J}\, F_{i,t}(Y) \;+\; \frac{J-12}{J}\, m^{\text{fwd}}_{i,t}(J-12), \tag{13}$$

where $m^{\text{fwd}}_{i,t}(J-12)$ is the average expected monthly inflation over the $J - 12$ months preceding the target year, itself identified from forecasts for earlier target years using the same procedure. Forecasts from Consensus Economics is reported on a calendar year basis (year ending in December), while forecasts from ESP are reported on a fiscal year basis (year ending in March). The conversion is applied separately within each event definition, and the resulting fixed-horizon observations are pooled only after conversion.

3.2.1 Forecast performance

Forecast errors are computed as the difference between realized CPI headline inflation of the appropriate event and the reported expectation, so that a negative forecast error implies that expected inflation exceeded realized inflation. Horizons below 3 months and above 120 months are excluded. Figure 3 reports the distribution of forecast errors across sources over time, and Figure 4 reports the

mean and variance of forecast errors by source and horizon, which serve as in-sample counterparts to the population forecast bias and MSE.

First, we observe that forecast error standard deviation declines with horizon across all sources (Figure 4b). To interpret this pattern, consider a simple unobserved components decomposition of inflation:

$$\pi_t = \tau_t + \varepsilon_t, \qquad \tau_t = \tau_{t-1} + \nu_t,$$

where $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ is a transitory disturbance and $\nu_t \sim N(0, \sigma_\nu^2)$ is a permanent shock to the trend. For an agent with complete information, the optimal forecast for all horizons is precisely τ_t, and the forecast error variance for the average annualized inflation rate over $[0, h]$ is

$$\text{MSE}(h) = \frac{\sigma_\varepsilon^2}{h} + \sigma_\nu^2 \cdot \frac{(h+1)(2h+1)}{6h},$$

where the first term is decreasing in h and the second is increasing.[10] A declining MSE across horizons implies that transitory shocks dominate variation in inflation expectations. The declining profile observed in Figure 4b is therefore consistent with permanent shocks being small relative to transitory ones, and with patterns documented for US inflation by Stock and Watson (2007).

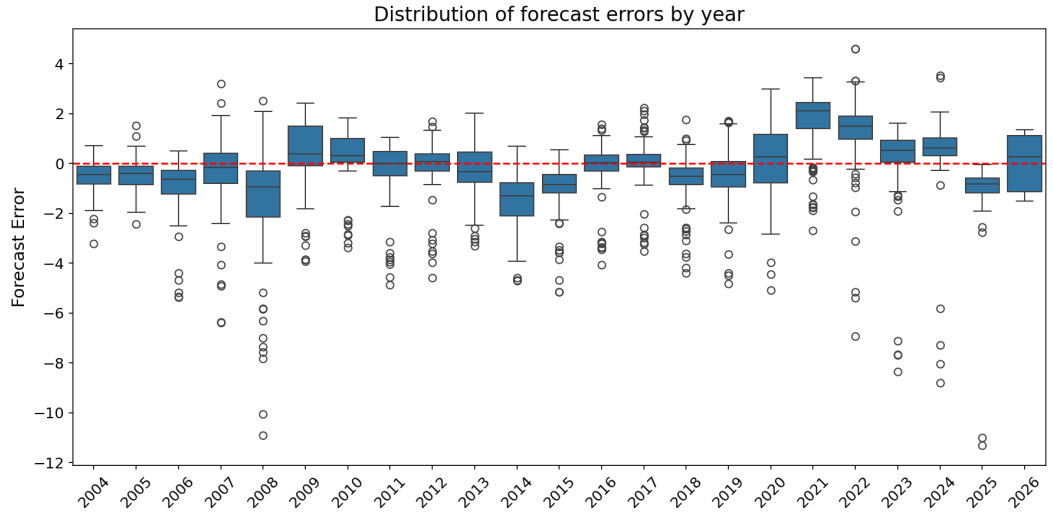
Second, professional survey sources (Consensus, ESP-A, and ESP-Q) exhibit small and relatively uniform biases across horizons, whereas household expectations display a pronounced positive bias (negative average forecast errors) at medium horizons (Figure 4a). This is consistent with the well-documented tendency of US households to be upwardly biased when forecasting inflation. Third, the distribution of forecast errors is broadly symmetric around zero in most years, but features notable negative outliers during the global financial crisis and in 2014, and a shift toward positive errors after 2021, reflecting the large and unexpectedly persistent inflation surprises of 2022–2023 (Figure 3a).

3.2.2 Forecast evaluation

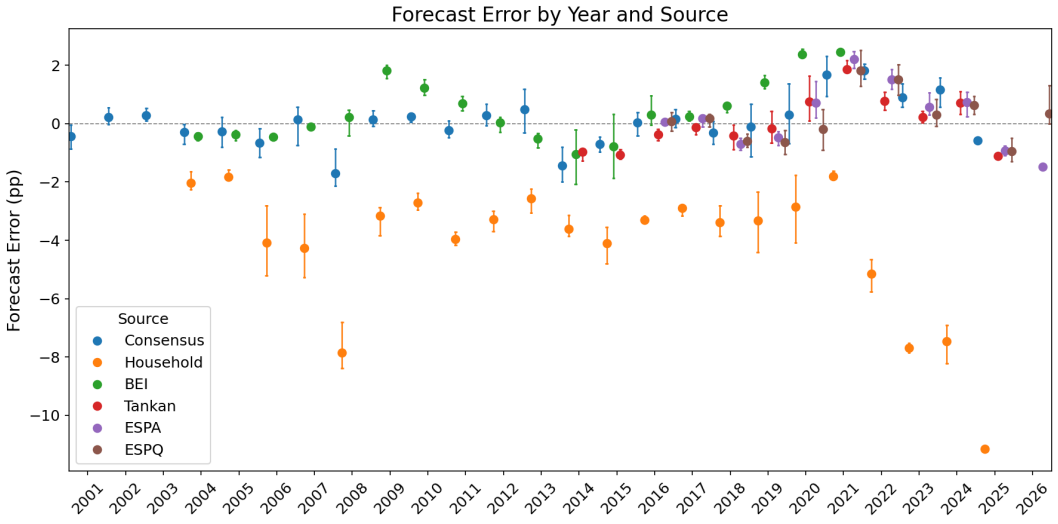
One challenge in interpreting in-sample MSE is sampling variation. In particular, comparisons are less reliable when sample coverage differs across sources or across forecast horizons. For example, 10-year forecasts can only be evaluated using realized inflation from up to 10 years ago, and therefore exclude the recent post-Covid pickup in inflation, whereas near-term expectations are evaluated up to almost the latest realizations. To facilitate comparison across sources, we instead consider performance relative to a common statistical benchmark, described in Section 3.1.5, evaluated over matched samples and horizons. While this mitigates differences in sample coverage and aids in comparisons, relative performance may still vary across evaluation windows so comparisons should be interpreted with caution.

A second challenge is that expectations are observed sparsely at specific horizons. To address

[10] The expression for complete-information MSE for the unobserved components model provides a lower bound on forecast MSE under incomplete information, where agents must additionally filter the unobserved trend from noisy inflation data.

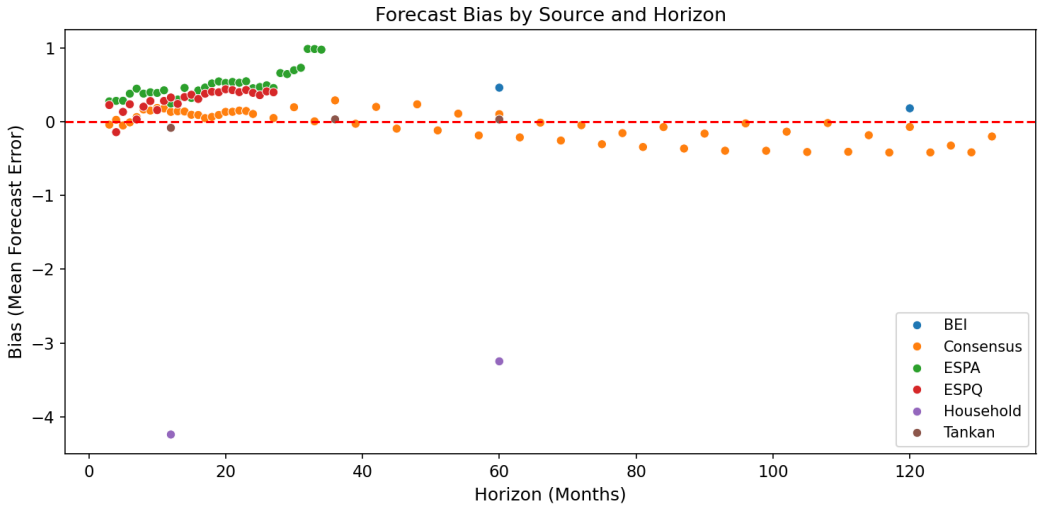


(a) Distribution of forecast errors by year

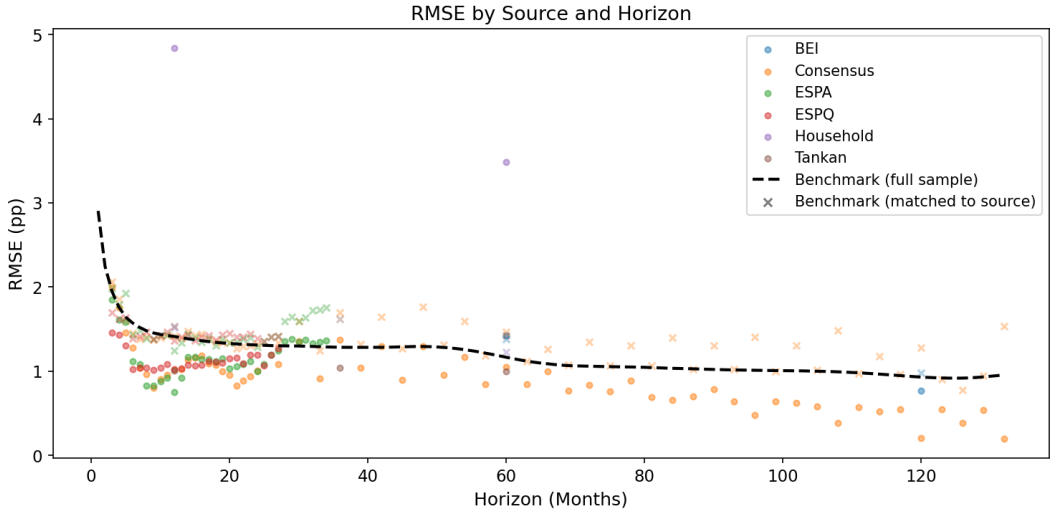


(b) Distribution of forecast errors by source and year

Figure 3: Forecast errors across sources and year. Panel (a) shows the distribution of forecast errors by year across all source-horizon pairs. Panel (b) separates the distribution of forecast performance by source and year.



(a) Mean forecast errors by horizon (months)



(b) Relative Mean Squared Errors (RMSE) of forecast errors by horizon (months)

Figure 4: Forecast performance across sources and horizons. Panels (a) and (b) show the bias and relative mean squared errors of forecast errors as a function of the forecast horizon in months. The benchmark statistical model is a simple rolling-mean forecast, described in Section 3.1.5.

this, we group observations into horizon bins (3–6, 7–12, 13–24, 25–36, 37–60, and 61–120 months). Not all sources span all bins, resulting in an unbalanced panel for the evaluation of sources across horizons.

Specifically, we report the relative root mean squared error at selected horizons, defined as

$$\text{Relative RMSE}_i(h) = \frac{\text{RMSE}_i(h)}{\text{RMSE}_{\text{BM}}(h)}, \tag{14}$$

where both the source and benchmark RMSE are computed over the exact same set of observations for source i within each horizon bin to ensure sample alignment. Table 1 reports these relative performance measures by source and binned forecast horizon.

Table 1: Relative RMSE by source and binned horizons. Each entry reports the ratio of the source's RMSE to the rolling-mean benchmark RMSE. Values below one (bold) indicate that the source outperforms the benchmark.

	Horizon (months)					
Source	3–6	7–12	13–24	25–36	37–60	61–120
Consensus	**0.927**	**0.664****	**0.747****	**0.789*****	**0.741**	**0.640*****
ESP-A	**0.860**	**0.624***	**0.785**	**0.809**	—	—
ESP-Q	**0.810****	**0.715***	**0.769**	**0.834*****	—	—
Tankan	—	**0.641***	—	**0.637**	**0.707*****	—
BEI	—	—	—	—	1.045	**0.784**
Household	—	3.150***	—	—	2.864	—

Notes: Stars denote rejection of the null of equal predictive accuracy in a two-sided Diebold-Mariano test of the source vs. rolling-mean benchmark squared forecast errors. Within each (source, bin) cell, observations are collapsed to origin-month frequency by averaging squared-error differentials across horizons in the bin. The long-run variance of the loss differential is estimated with a Newey-West Bartlett kernel and bandwidth equal to the median forecast horizon (in months) within the bin. The test statistic is evaluated against t_{N-1} critical values after the Harvey-Leybourne-Newbold (1997) small-sample correction. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Several patterns emerge. First, all professional forecaster sources, Consensus, ESP-A, and ESP-Q, outperform the benchmark at every reported horizon, with relative RMSEs generally in the range 0.6–0.9. Across horizons, the degree of outperformance varies systematically, with a sharp improvement from very short to one-year horizons, relatively stable performance at medium horizons, and some further improvement at long horizons. Second, Tankan firm surveys also display strong outperformance, though these are forward-window forecasts of $[h-12, h]$ and are therefore not directly comparable, as they omit the $[0, h-12]$ window, which is generally harder to forecast. Third, 5-year BEI underperforms the benchmark, potentially reflecting maturity-dependent premia embedded in market-based measures, while 10-year BEI outperforms the benchmark but still underperforms Consensus at comparable horizons. Fourth, household expectations underperform the benchmark by a wide margin, reflecting the persistent upward bias documented previously.

3.3 Source Commonality and Heterogeneity

To characterize the extent of comovement in inflation expectations across sources and horizons, we conduct a principal components analysis (PCA) on a panel of expectations indexed by time and by source-horizon pairs. Expectations are grouped into horizon bins (3–6, 7–12, 13–24, 25–36, 37–60, 61–120 months), and observations are averaged within each (source, bin, quarter) cell. The analysis is restricted to 2016Q1–2025Q4, yielding a balanced panel of 16 source-bin series, see Table 2. All series are standardized prior to estimation to remove differences in scale across sources.

Table 2: Source-horizon bin series included in the PCA panel.

Source	Horizon bin (months)	Description
BEI	37–60	Market-implied breakeven inflation (5-year)
BEI	61–120	Market-implied breakeven inflation (10-year)
Consensus	13–24	Professional forecasters (1–2 years)
Consensus	25–36	Professional forecasters (2–3 years)
Consensus	37–60	Professional forecasters (3–5 years)
Consensus	61–120	Professional forecasters (5–10 years)
ESP-A	13–24	ESP annual survey (1–2 years)
ESP-A	25–36	ESP annual survey (2–3 years)
ESP-Q	3–6	ESP quarterly survey (0–6 months)
ESP-Q	7–12	ESP quarterly survey (6–12 months)
ESP-Q	13–24	ESP quarterly survey (1–2 years)
Household	7–12	BoJ household survey (1 year)
Household	37–60	BoJ household survey (5 years)
Tankan	7–12	BoJ Tankan survey (1 year)
Tankan	25–36	BoJ Tankan survey (3 years)
Tankan	37–60	BoJ Tankan survey (5 years)

Notes: Four initial quarters of missing observations for ESPA in the 25–36 month horizon bin are imputed using an EM-PCA procedure, which iteratively fills missing entries using the current low-rank reconstruction (E-step) and refits the principal components (M-step) until convergence.

Table 3: Explained variance by principal component.

Component	Variance share	Cumulative	Interpretation
PC1	91.4%	91.4%	Common level factor
PC2	4.7%	96.1%	Slope and curvature
PC3	2.4%	98.5%	Source disagreement

Notes: The PCA is estimated on a balanced panel of 16 source-horizon series over 2016Q1–2025Q4, constructed by averaging expectations within horizon bins (3–6, 7–12, 13–24, 25–36, 37–60, 61–120 months).

Table 3 reports the variance decomposition from the PCA. The first principal component explains 91.4 percent of total variance and exhibits near-uniform loadings across all source-horizon pairs. This indicates the presence of a dominant common level factor that accounts for the vast majority of variation in inflation expectations across sources and horizons. Meanwhile, the second principal component explains an additional 4.7 percent of the variance, so that the first two

components together account for over 96 percent of the total variation, while the third component accounts for only an additional 2.4 percent.

The loadings of the second principal component exhibit a monotone pattern across horizons, with negative weights at short horizons and positive weights at long horizons, consistent with capturing the variation in expectations of a transitory shock. In contrast, the third component explains only a small additional share of variation and does not display a systematic term-structure pattern; instead, its loadings are organized by source, with household and market-based measures loading positively and professional forecaster surveys loading negatively. This indicates that the third component primarily captures cross-source disagreement rather than a distinct term-structure factor. Taken together, these results are consistent with a single common underlying term structure across sources, with residual source-specific variation playing a limited role.

3.3.1 Expectation term structure and Nelson-Siegel loadings

To isolate term-structure variation from cross-source disagreement, we repeat the analysis on a panel constructed by averaging expectations across sources within each horizon bin. In this collapsed 6-series panel, the first two principal components account for 98.0 percent of the variance, leaving very little residual variation that is unexplained.

This contrasts from the yield curve literature which typically find at least three term-structure factors. The apparent two-factor structure in inflation expectations, however, does not necessarily imply that a three-factor NS representation is inappropriate. First, the use of coarse horizon bins limits the cross-sectional resolution. This is particularly relevant at the short end, where slope and curvature loadings differ most, making it difficult to distinguish the two components empirically for our sample. Second, at quarterly frequency, the underlying slope and curvature factors may exhibit substantial co-movement over time (as we find in our application), so that PCA recovers a single combined component rather than two separate directions. As a result, the second principal component may instead be reflecting a blend of slope and curvature loadings.

To examine this hypothesis, we next assess whether the cross-sectional variation in the data is consistent with the NS loading structure, which underlies our NSICX model. The NS framework posits three factors with fixed loading shapes across horizons, governed by a decay parameter λ. To evaluate this loading restriction on the term structure, we construct the NS loading matrix with $\lambda = 0.16$ (consistent with our application) at the six horizon-bin midpoints and project the standardized data onto this basis via least squares. Specifically, letting Y_t denote the 6×1 vector of standardized expectations across bins at time t, and $B(\lambda)$ the 6×3 matrix of NS loadings (level, slope, curvature), we estimate the regression

$$Y_t = B(\lambda) f_t + u_t,$$

where f_t are time-varying factor scores obtained by OLS, and u_t is a residual. The fitted values are then used to compute the pooled R^2 across all series and time periods. We compare the fit of the NS representation to that of an unconstrained PCA.

Table 4: Fit of Nelson-Siegel representation relative to PCA benchmark.

Model	R^2	Incremental gain	p-value
Nelson-Siegel (constrained)			
Level only	95.18%	—	—
Level + slope	97.42%	+2.24pp	< 0.001
Level + slope + curvature	97.85%	+0.43pp	≈ 0.01
Free PCA (unconstrained)			
1 component	95.20%	—	—
2 components	97.98%	+2.78pp	—

Table 4 reports the R^2 from successive specifications as factors are added, alongside the corresponding unconstrained PCA benchmarks. The level factor alone explains just over 95 percent of the variation, closely matching the explanatory power of the first principal component estimated on the 6-series panel. Adding the slope factor leads to a large and statistically significant improvement in fit, while the curvature factor provides a smaller but still statistically significant incremental gain, bringing the overall fit of the three-factor NS model in line with that of the two-component PCA.

Moreover, we find that the fit of the three-factor NS model is essentially invariant to λ in a neighborhood of the calibrated value. For $\lambda \in [0.06, 0.2]$, the three-factor R^2 varies only between 97.77% and 98.02%. This suggests that the cross-sectional variation in expectations alone provides weak identification of the curvature peak and the slope decay rate. In contrast, the NSICX estimation additionally exploits the transition dynamics of the latent factors to identify λ, which is likely to be particularly informative in this setting.

To summarize, we find that the explanatory power of the second PCA component closely corresponds to the combined contribution of slope and curvature in the NS representation, supporting the interpretation that these two factors are jointly captured by a single blended second principal component. Overall, these results support the use of a common latent term structure for inflation expectations across sources, well captured by the low-dimensional NS representation.

4 Results

We estimate the NSICX model on the June 2026 monthly sample, which spans inflation expectations from January 2004 through June 2026 using the six data sources described in Section 3.[11] The first 12 months are used for filter initialization and are excluded from the reported estimates, so results are reported from January 2004 onward. Also described in Section 3, we apply the calendar-year conversion to all Consensus Economics and ESP fixed-event forecasts, and assign Tankan observations the forward-average loadings over $[h-12, h]$ months. Observations with horizons below 3 months or above 120 months are excluded. As described in Section 2.4, the shape parameter λ and transition innovation variances (q_L, q_S, q_C) are estimated jointly by MLE, while measurement

[11] All expectation inputs in this sample were publicly available by July 16, 2026, when the Bank of Japan released the June round of the Opinion Survey, the last of the six sources to be published.

error variances are held fixed at their calibrated values.

4.1 Estimation

Table 5 reports MLE estimates of the four free parameters. The level innovation variance q_L and the decay parameter λ are statistically significant. The slope innovation variance q_S is estimated at zero, indicating no direct innovations to the slope factor, with slope instead driven indirectly by the curvature factor. The estimated λ of 0.1601 is slightly higher than the US estimate of 0.121 from Aruoba (2020), implying faster decay in the slope component and an earlier curvature peak, at $1/0.1601 \approx 6.2$ months versus $1/0.121 \approx 8.3$ months for the US.

Table 5: Maximum Likelihood Parameter Estimates

Parameter	Estimate	Std. Error	95% Lower	95% Upper
q_L	0.0065	0.0019	0.0027	0.0103
q_S	0.0000	0.2082	−0.4081	0.4081
q_C	0.2577	0.3102	−0.3503	0.8657
λ	0.1601	0.0753	0.0125	0.3078

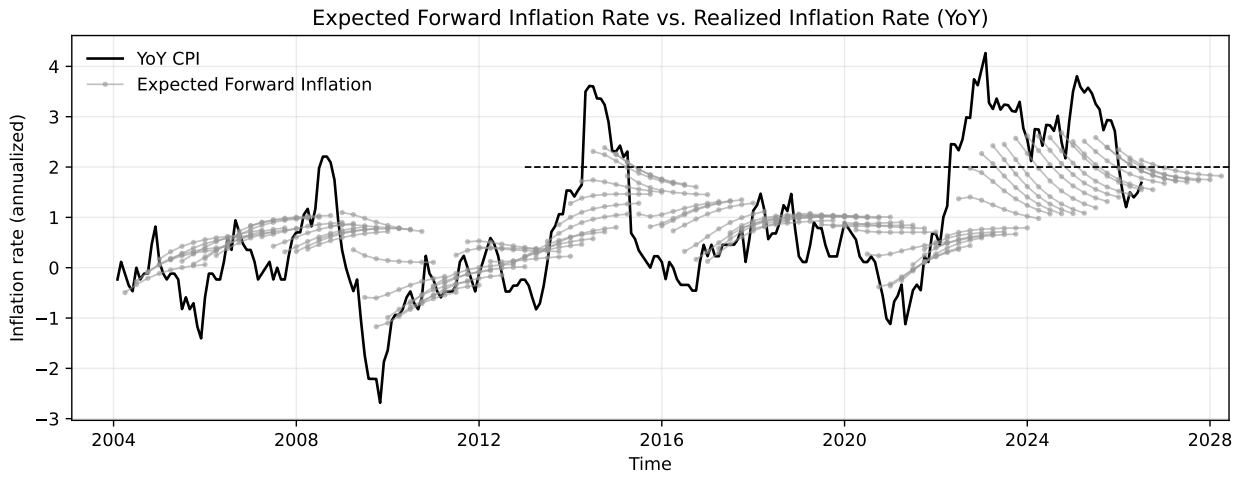
Notes: $T = 282$ monthly observations. Standard errors computed via the outer product of gradients (OPG). Parameters q_L, q_S, q_C are the diagonal elements of the state innovation covariance matrix Q; λ is the Nelson-Siegel decay parameter.

Before turning to the central estimates and their uncertainty, we first illustrate the model's output in two ways that highlight fit and cross-sectional shape (Figure 5).

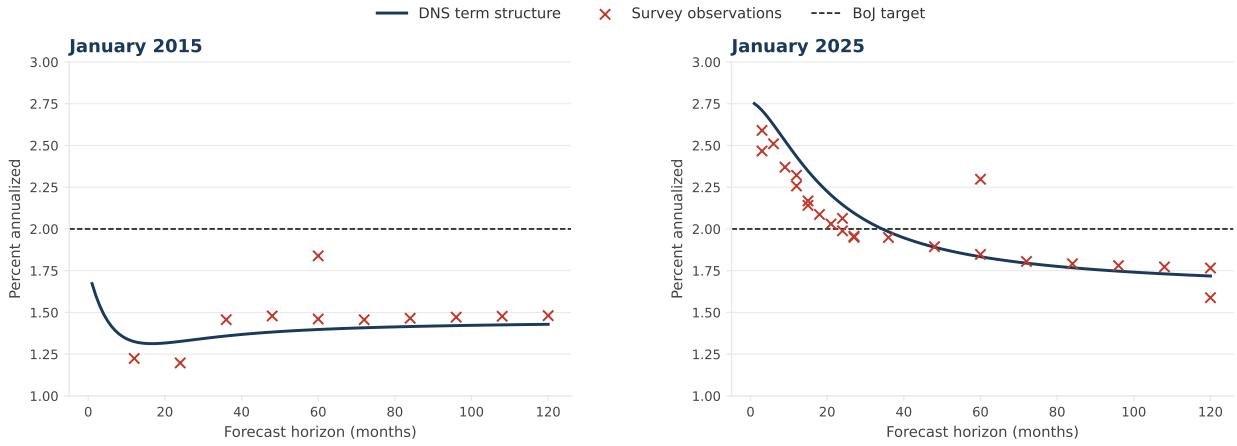
Figure 5a plots real-time NSICX inflation expectations alongside realized CPI inflation. For every third survey origin, a grey trajectory ("whisker") traces the instantaneous forward inflation rate at horizons $h = 3, 6, 9, \ldots, 24$ months ahead.[12] Each whisker represents a real-time snapshot of the expected inflation path over the subsequent two years, originating near the prevailing CPI inflation rate and tracing the expected convergence path toward the trend inflation expectation, which itself varies over time. Trend inflation rose steadily in the pre-GFC period to approximately 1 percent around 2007, declined sharply to a negative trough around 2010, recovered to around 1.5 percent by 2015, drifted down through the pre-pandemic years to just above 0.5 percent, and has since risen markedly to the range of 1.5 to 2 percent. This recent range is broadly consistent with the BOJ's 2-percent price stability target, adopted in January 2013.[4]

Figure 5b shows cross-sectional snapshots of the smoothed NSICX term structure alongside survey observations for January 2015 and January 2025, from which three features emerge. First, the estimated term structure fits the measurements well in both periods. Second, the density of observations is greater in 2025, reflecting the expansion of survey sources. Third, in both snapshots, household survey expectations at the five-year horizon stand out as upside outliers relative to professional, firm, and market-based measures.

[12]The expected instantaneous forward inflation rate is computed from the filtered DNS factors as $\hat{\pi}_{t,h} = L_t + \mathcal{L}_S(h;\lambda)S_t + \mathcal{L}_C(h;\lambda)C_t$, where $\mathcal{L}_S(h;\lambda)$ and $\mathcal{L}_C(h;\lambda)$ are the NS forward loadings at horizon h.



(a) Real-time forward-rate trajectories. Grey trajectories show instantaneous forward rates at horizons $h = 3, 6, \ldots, 24$ months ahead for every third survey origin. The black line is realized year-on-year CPI inflation. The dashed line marks the BOJ 2-percent price stability target, which the Bank of Japan introduced in January 2013.



(b) Smoothed NSICX term structure and survey observations at January 2015 (left) and January 2025 (right).

Figure 5: NSICX term structure fit and cross-sectional shape.

4.2 Central Estimates and Uncertainty

Figure 6 reports central estimates of the NSICX term structure together with pointwise 68- and 95-percent uncertainty bands.

Figure 6a plots the filtered and smoothed estimates of the three latent DNS factors over the full sample. The two series track each other closely, with the smoothed estimates exhibiting somewhat less noise. The level factor L_t captures the trend rate of inflation expectations described above; the slope and curvature factors carry the deviations from trend at short and medium horizons.

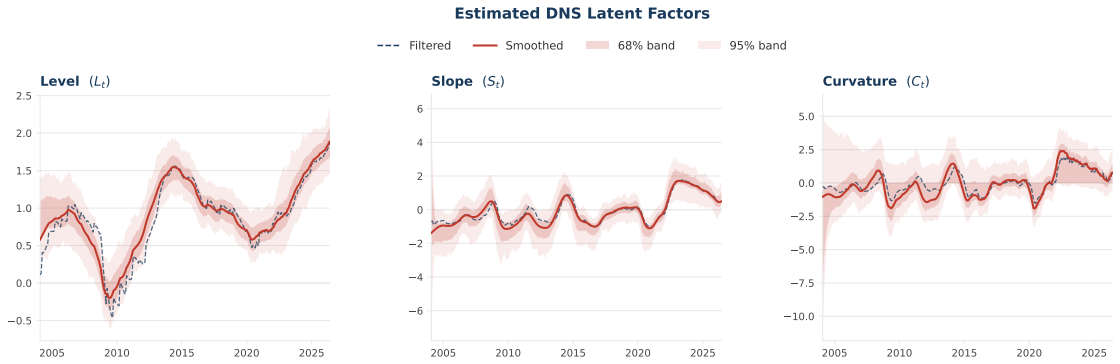
Figure 6b shows the term structure at the end of the sample. The point estimate of the term structure tilts mildly downward, from above the 2-percent target at short- and medium-term horizons to marginally below it at longer horizons (with the target sitting within the 95-percent confidence bands throughout). The dashed red line is the MLE estimate, which aligns closely with the posterior median, suggesting limited impact of parameter uncertainty on the central estimate, while the confidence bands across horizons reflect joint uncertainty in (L_t, S_t, C_t). This shape suggests that the rise in energy prices in the wake of the Iran war continues to be perceived as a largely transitory shock with limited pass-through to long-run expectations.

Figure 6c plots the time series of trend inflation L_t and the 2-year expectation $\hat{\pi}_{t,24}$ over the full sample, the two summary statistics most relevant for assessing alignment with the BOJ target. Both central lines are the posterior median of the smoothed (full-sample) state estimate, taken across the simulation-smoother draws, so that the plotted line is consistent with its surrounding confidence bands.[13] The two series track each other closely, reflecting the rapid decay of the slope and curvature loadings, with differences most apparent during periods of sharp inflation movements when contributions from the slope and curvature components are more pronounced. The BOJ target enters the 95-percent confidence band for the 2-year expectation by mid-2022 and for L_t by late 2024; the central estimate of the 2-year expectation reaches the target by early 2024. This pattern is broadly consistent with inflation expectations aligning with the BOJ target in levels by the start of the hiking cycle in March 2024.

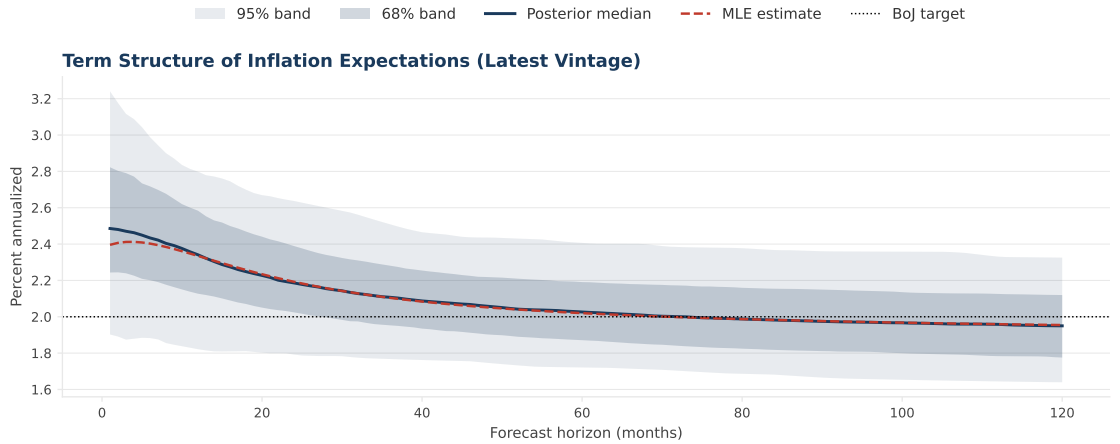
4.3 Forecasting Performance

Finally, we report the out-of-sample forecasting performance of our real-time inflation expectations. Table 6 reports the relative RMSFE of real-time NSICX composite expectations against a rolling-mean benchmark at selected horizons. The composite outperforms the benchmark at all horizons, with relative RMSFEs of 0.74–0.77 at short to medium horizons and 0.48 at the 10-year horizon. The outperformance relative to the statistical benchmark is statistically significant at horizons up to 36 months, but not for 60 and 120 months. The latter could be partly driven by a smaller sample size and greater correlation in forecast errors due to the overlapping evaluation windows. Its out-of-sample performance is broadly comparable to Consensus Forecasts at short- and medium-term horizons, but it outperforms Consensus at long horizons (5–10 years).

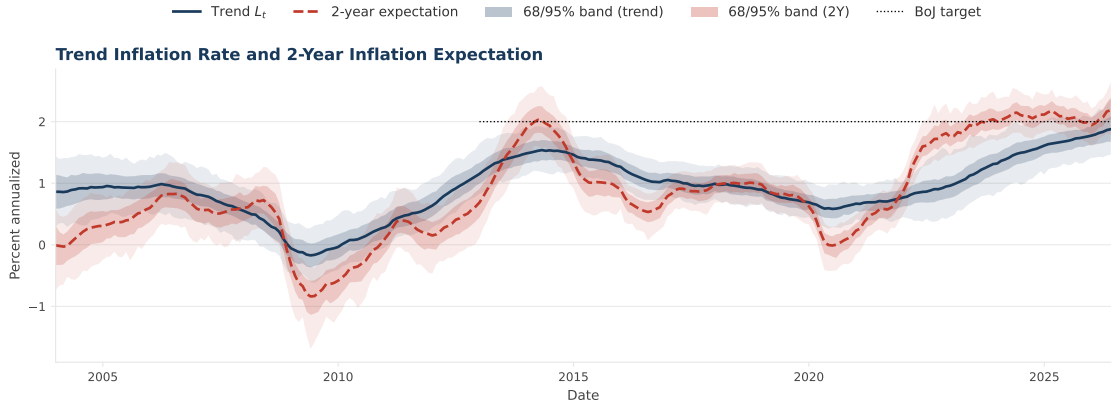
[13] The 2-year expectation is computed as $\hat{\pi}_{t,24} = L_t + \mathcal{L}_S(24;\hat{\lambda})S_t + \mathcal{L}_C(24;\hat{\lambda})C_t$, where $\mathcal{L}_S$ and $\mathcal{L}_C$ are the Nelson-Siegel loadings.



(a) Filtered and smoothed estimates of the three DNS latent factors, level (L_t), slope (S_t), and curvature (C_t). Shaded bands show pointwise 68- and 95-percent intervals around the smoothed estimate.



(b) Smoothed estimate of the term structure, June 2026 sample. Blue line: posterior median; shaded bands: 68- and 95-percent intervals; dashed red line: MLE estimate.



(c) Trend inflation L_t and 2-year expectation $\hat{\pi}_{t,24}$, each plotted as the posterior median of the smoothed (full-sample) state estimate across the simulation-smoother draws. Blue: L_t; red dashed: $\hat{\pi}_{t,24}$. Shaded bands: pointwise 68- and 95-percent intervals. Dotted line: BOJ 2-percent target, drawn from January 2013 onward, when the Bank of Japan adopted the 2-percent price stability target. L_t corresponds to the limiting value of the forward rate as $h \to \infty$.

Figure 6: NSICX central estimates and uncertainty.

Table 6: Relative RMSE of the NSICX expectations against the statistical benchmark at selected horizons since 2002. Values below one indicate that the composite outperforms the benchmark.

	Horizon (months)					
	6	12	24	36	60	120
Evaluation sample end	2026-01	2025-07	2024-07	2023-07	2021-07	2016-07
N	265	259	247	235	211	151
RMSFE benchmark (pp)	1.734	1.555	1.425	1.385	1.209	0.972
RMSFE NSICX (pp)	1.335	1.148	1.053	1.029	0.927	0.468
Relative RMSFE	**0.770*****	**0.738*****	**0.739****	**0.743****	**0.767**	**0.481**

Notes: Stars denote rejection of the null of equal predictive accuracy in a two-sided Diebold-Mariano test of NSICX vs. benchmark squared forecast errors. The long-run variance of the loss differential is estimated with a Newey-West Bartlett kernel and bandwidth equal to the forecast horizon h (in months). The test statistic is evaluated against t_{N-1} critical values after the Harvey-Leybourne-Newbold (1997) small-sample correction. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

The alignment in levels documented in Section 4 does not by itself establish that the anchor is firmly in place, which would require expectations to be insensitive to data outturns, a possibility we explore in Section 4.4.

4.4 Long-term Inflation Expectation in Japan vs the US

We next examine the time series variation in long-term inflation expectations in Japan and the US. Figure 7 compares the model-implied trend inflation rate L_t from our NSICX model for Japan with the analogous level factor for the US from the June 2026 vintage of the ATSIX model. Both series are filtered trend components of a DNS term structure of inflation expectations, but given that they are estimated using different models, the following comparison reflects differences in model specification as well as genuine differences in inflation expectations between Japan and the US.

The two series display a striking divergence. US trend inflation remained close to or above 2 percent throughout, and was stable during both the post-GFC and post-pandemic periods. Japan's trend, by contrast, exhibits persistent shifts over recent decades, remaining below 2 percent for most of the sample and rising toward the 2-percent target only near the end.[4]

Table 7 formalizes this contrast by examining the volatility of changes in trend inflation expectations. Under a random walk specification, the standard deviation of ΔL_t measures month-to-month short-term volatility, while that of $L_t - L_{t-12}$ captures persistence. If innovations are serially uncorrelated in accordance with our transition model, the ratio of 12-month to month-to-month volatility should equal $\sqrt{12} \approx 3.46$. This provides a natural benchmark for comparing both the volatility and persistence of changes in trend inflation expectations, as ratios below this indicate some degree of mean reversion or stationarity, while ratios at or above it indicate unit-root behavior. We formalize this comparison with a variance ratio test in the spirit of Lo and MacKinlay (1988). Defining $VR(12) = \mathrm{Var}(L_t - L_{t-12}) / (12\,\mathrm{Var}(\Delta L_t))$, the random walk null implies $VR(12) = 1$, while anchored (stationary) expectations imply $VR(12) < 1$. We therefore test the unit-root null against the one-sided anchored alternative, with p-values computed using the wild bootstrap of

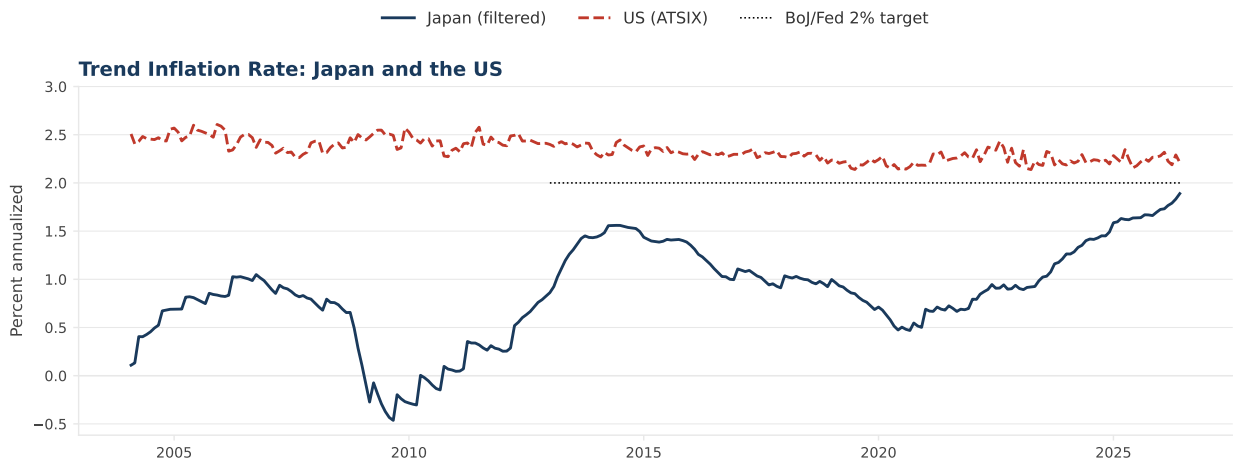


Figure 7: Model-implied trend inflation for Japan and the US. The solid blue line shows the filtered estimate of the NSICX level factor L_t for Japan. The dashed red line shows the level factor from the ATSIX model. The dotted line marks the 2-percent target; the Bank of Japan adopted the 2-percent price stability target in January 2013, and the Federal Reserve adopted its 2-percent longer-run inflation objective in January 2012.

Kim (2006), which is robust to conditional heteroskedasticity.[14]

Table 7: Variability of Trend Inflation Rate L_t: Japan and the US

	Japan	US	Benchmark
Std. dev., 1-month difference (percent)	6.416	5.891	$\sqrt{q_L}$
Std. dev., 12-month difference (percent)	33.673	8.467	$\sqrt{12q_L}$
Ratio (12-month / 1-month)	5.25	1.44	3.46
Variance ratio $VR(12)$	2.30	0.17	1
p-value (one-sided, $VR < 1$)	0.997	< 0.001	—

Notes: Standard deviations of first differences of the filtered DNS level factor L_t for Japan and the ATSIX level factor for the US, computed over the common sample. $VR(12)$ is the variance ratio $\mathrm{Var}(L_t - L_{t-12})/(12\,\mathrm{Var}(\Delta L_t))$, equal to 1 under the random walk null. p-values test the unit-root null against the one-sided anchored alternative $VR < 1$, via wild bootstrap (10,000 replications).

Short-run volatility is similar across countries, with standard deviations of 0.064 and 0.059 percentage points for Japan and the US.[15] At the 12-month horizon, however, the patterns diverge. The US ratio of 1.44 corresponds to $VR(12) = 0.17$, and the test rejects the unit-root null in favor of the anchored alternative ($p < 0.001$), consistent with mean reversion and strong anchoring.[16]

[14] Each of 10,000 bootstrap replications resamples the demeaned first differences of L_t with Rademacher weights and cumulates them under the null; the p-value is the share of bootstrap variance ratios at or below the sample estimate.

[15] As a consistency check, $\sqrt{q_L} = \sqrt{0.0065} \approx 0.081$ percent per month is close to the empirical standard deviation of ΔL_t of 0.064 percent, with differences reflecting filtering noise.

[16] While the ATSIX AR(3) transition model is not internally consistent with the NS loading structure, it can still fit US inflation expectations well in-sample due to strong anchoring, though the implied volatility depends on the assumed transition dynamics.

Note that part of this contrast may reflect model structure, since the stationary AR(3) transition model in ATSIX dampens long-horizon variation in the filtered US level factor by construction, but in the case of the US the stability of long-run expectations is well established outside of the context of ATSIX. Japan's ratio of 5.25 corresponds to $VR(12) = 2.30$, and the test fails to reject the unit-root null ($p = 0.997$), consistent with the sustained shifts in Figure 7. With this caveat in mind, these results also inform the appropriate specification of the level factor: a stationary level, as assumed in ATSIX, is supported for the US, while the Japanese data favor the unit-root level factor implied by our internally consistent transition equation.

Next, we formally test for anchoring by regressing quarterly innovations in L_t on proxies for short-run inflation information, following Crump and others (2023), Gürkaynak and others (2010) and Bems and others (2021): realized quarter-on-quarter annualized CPI inflation π_t^{qoq}, year-on-year CPI inflation π_t^{yoy}, the deviation of inflation from trend $\pi_t^{qoq} - L_{t-1}$, and the fixed-event revision in short-term expectations derived from the NS term structure.[17]

The regression takes the form

$$\Delta L_t = \alpha + \beta x_t + \varepsilon_t, \tag{15}$$

where a statistically significant β indicates that long-run expectations respond systematically to short-run information, which we interpret as evidence of imperfect anchoring. We estimate (15) with HAC standard errors over the common sample from 2004Q2 to 2026Q2, using the ATSIX level factor as the US counterpart to L_t. The regression is estimated at the quarterly frequency rather than monthly to avoid timing complications arising from misalignment between CPI releases and survey dates due to reporting lags and sampling variation. Table 8 reports the results.

Table 8: Long-term inflation expectation anchoring regressions

	Japan (NSICX)				US (ATSIX)			
x_t	$\hat{\alpha}$	$\hat{\beta}$	R^2	p-val	$\hat{\alpha}$	$\hat{\beta}$	R^2	p-val
π_t^{qoq}	0.007	0.016*	0.064	0.100	−0.001	0.000	0.000	0.868
π_t^{yoy}	0.015	0.007	0.005	0.481	−0.005	0.001	0.001	0.716
$\pi_t^{qoq} - L_{t-1}$	0.021	0.021**	0.086	0.043	−0.002	0.000	0.000	0.952
Δ_t^{ST}	0.015	0.320***	0.387	0.000	−0.004	0.073***	0.135	0.002

Notes: $T = 89$ quarterly observations, 2004Q2–2026Q2. HAC standard errors with 4 lags.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Δ_t^{ST} denotes the fixed-event short-term revision.

For the US, the level factor L_t from the ATSIX model shows no statistically significant response to realized inflation measures. Only the fixed-event short-term revision Δ_t^{ST} yields a positive and significant coefficient of 0.073 with an R^2 of 0.14, indicating that short-run expectation dynamics explain a modest share of variation in the US trend, and broadly consistent with well-anchored

[17] The fixed-event revision at the short end is defined as

$$\Delta_t^{\mathrm{ST}} = \hat{\pi}_t(0,3) - \hat{\pi}_{t-1}(3,6),$$

where $\hat{\pi}_t(h_1,h_2) = L_t + \mathcal{L}_S(h_1,h_2;\lambda)S_t + \mathcal{L}_C(h_1,h_2;\lambda)C_t$.

long-run expectations over the sample.[18]

For Japan, long-run expectations appear less well anchored on average over the sample. Realized inflation enters positively under the quarter-on-quarter specification ($\hat{\beta} = 0.016$, $p = 0.10$), and with a larger coefficient under the deviation from trend specification ($\hat{\beta} = 0.021$, $p = 0.04$). The fixed-event short-term revision has the strongest explanatory power ($\hat{\beta} = 0.320$, $p < 0.001$, $R^2 = 0.39$), indicating that near-term expectation revisions are highly predictive of changes in the perceived trend. The coefficient estimates indicate that, over the sample period, the pass-through of variations in short-run inflation expectation to long-run expectations is roughly four times larger in Japan.

Figure 8 re-estimates equation (15) on a 20-quarter rolling window using Δ_t^{ST} as the regressor. Japan's coefficient hovered around 0.5 through early 2014, above the unconditional estimate of 0.3 and roughly five times the contemporaneous US estimates. The magnitude of the estimate suggests a 50 percent pass-through of short-term shocks to inflation to long-run expectations. Following the introduction of Quantitative and Qualitative Easing (QQE) in April 2013, Japan's coefficient began stabilizing in the 0.15–0.25 range through the late 2010s. Interestingly, the coefficient for the US has drifted upwards during the period corresponding to quantitative easing in the US. Since the start of normalization in 2024, the coefficient for Japan has moved further down to approximately 0.08 by the end of the sample, broadly in line with the contemporaneous US estimate of 0.04. Taken together with the level convergence documented in Section 4, the rolling evidence is consistent with an alignment of long-run expectations in Japan with the BOJ's 2-percent target from below.

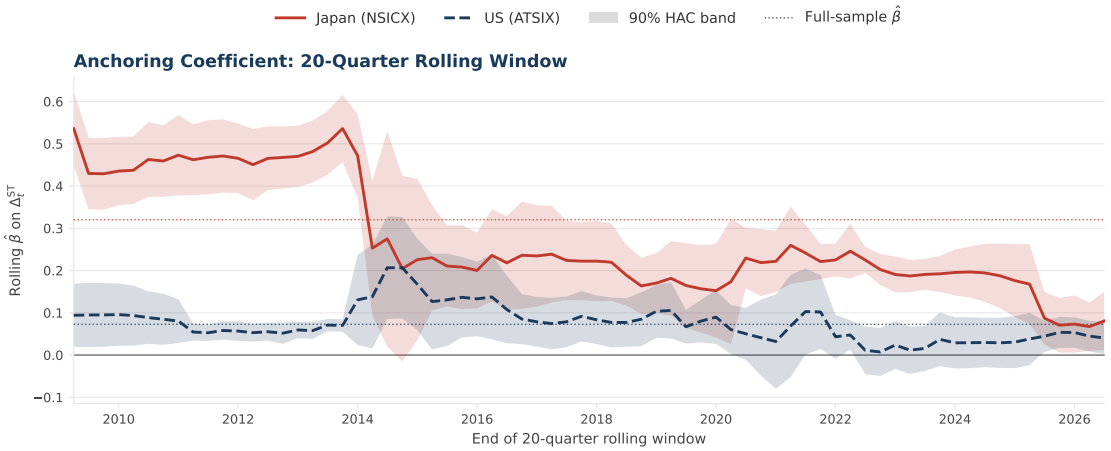


Figure 8: Rolling-window estimates of the anchoring coefficient β in $\Delta L_t = \alpha + \beta \Delta_t^{\mathrm{ST}} + \varepsilon_t$, estimated on a 20-quarter (5-year) right-aligned trailing window for Japan (NSICX, red solid) and the US (ATSIX, blue dashed). Shaded bands show 90-percent HAC confidence intervals (4 lags). Horizontal dotted lines mark the full-sample point estimates from Table 8.

[18] This residual pass-through aligns with Giacomini and others (2025), who show that US inflation expectations are generally well anchored when assessed through the sensitivity of long-term revisions to perceived shocks, but exhibit substantial time variation in perceived persistence; in our constant-parameter setting, this variation is averaged out over the sample.

4.5 Ex-ante Real Interest Rates and JGBi Yields

We next use our NSICX composite to construct ex-ante real interest rates. Under the Fisher decomposition, the ex-ante real rate at horizon h is

$$r_t(h) = i_t(h) - \hat{\pi}_t^e(h), \tag{16}$$

where $i_t(h)$ is the nominal JGB yield at maturity h and $\hat{\pi}_t^e(h)$ is the inflation expectation at the same horizon.

Meanwhile, BEI reflects not only expected inflation but also an inflation risk premium (IRP), a liquidity premium (LP), and a deflation protection (DP) component.[19] As a result, JGBi-implied real yields differ systematically from our composite-implied ex-ante real rates, with the wedge providing a reduced-form measure of these components. Because the ex-ante real rate inherits the estimation uncertainty of the NSICX term structure, we also include 95-percent confidence bands derived from the simulation-smoother draws described in Section 2.4, and focus our interpretation on periods when the wedge is statistically different from zero.[20] Figure 9a compares NSICX-implied ex-ante real rates with JGBi real yields at the 5-year ($h = 60$ months) and 10-year ($h = 120$ months) horizons from 2008 and 2004 onward, respectively.

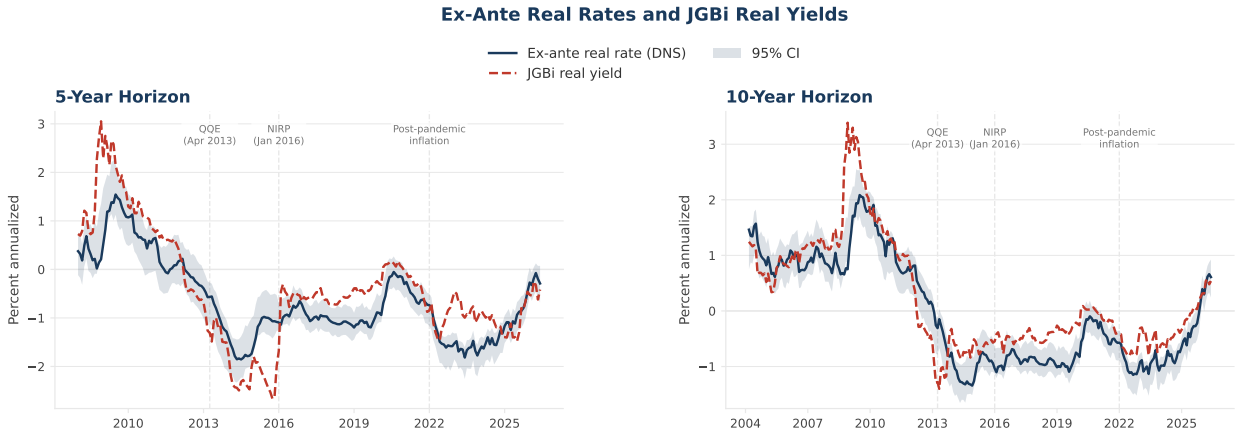
Figure 9a shows that JGBi real yields generally lie below NSICX-implied ex-ante real rates, implying a negative wedge for most of the sample, statistically different from zero at the 5-percent level in about 60 percent of months at both horizons. The wedge narrows and briefly turns positive around QQE in April 2013, consistent with compressed liquidity and risk premia, and widens sharply around NIRP in January 2016. Post 2022, the wedge narrows again as inflation expectations rise alongside nominal yields, and by the end of the sample it is mostly indistinguishable from zero at both horizons. These dynamics are broadly similar across horizons, although the 5-year wedge is more volatile.

Our findings are broadly consistent with the negative inflation risk premium documented by Christensen and Spiegel (2024), hereafter denoted CS. Figure 9b overlays the CS decomposition of BEI into IRP, LP, and DP. While the CS estimates are latent model-based measures fitted to JGB and JGBi prices, the two series are qualitatively similar, sharing the same sign for most of the sample and aligning at key turning points. However, the CS wedge exhibits larger swings, particularly at the 10-year horizon during 2016–2019. The largest divergence occurs in 2013–2016, when CS estimate a strongly negative LP that our survey-anchored expectations do not absorb to the same extent.

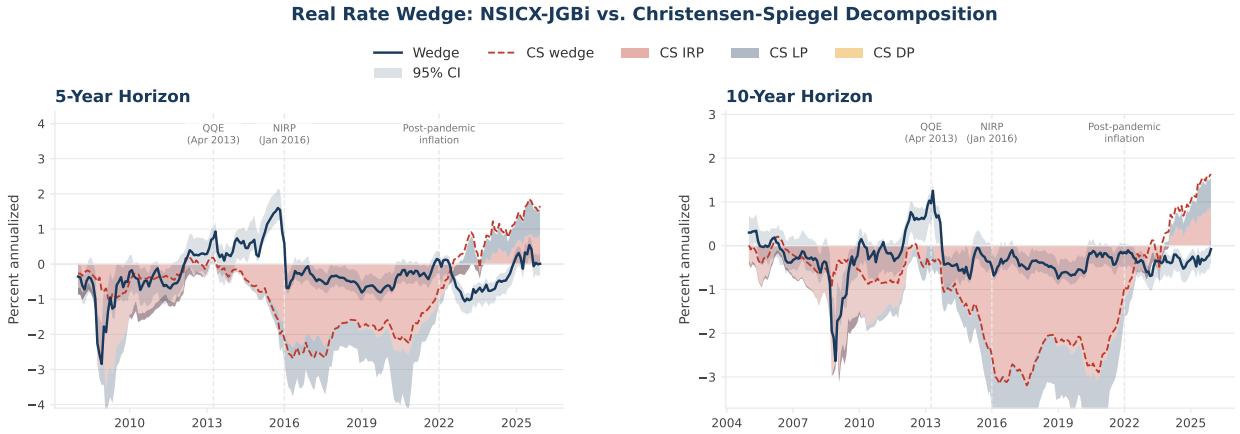
Figure 9c extends the analysis beyond long-term maturities to the 1- and 2-year horizons. Over the longer run, real rates fell sharply after the introduction of QQE. Prior to QQE, real rates were positive across all horizons, with the 10-year around 1–2 percent and shorter horizons near 0–0.5

[19] See Christensen and Spiegel (2024) for more details.

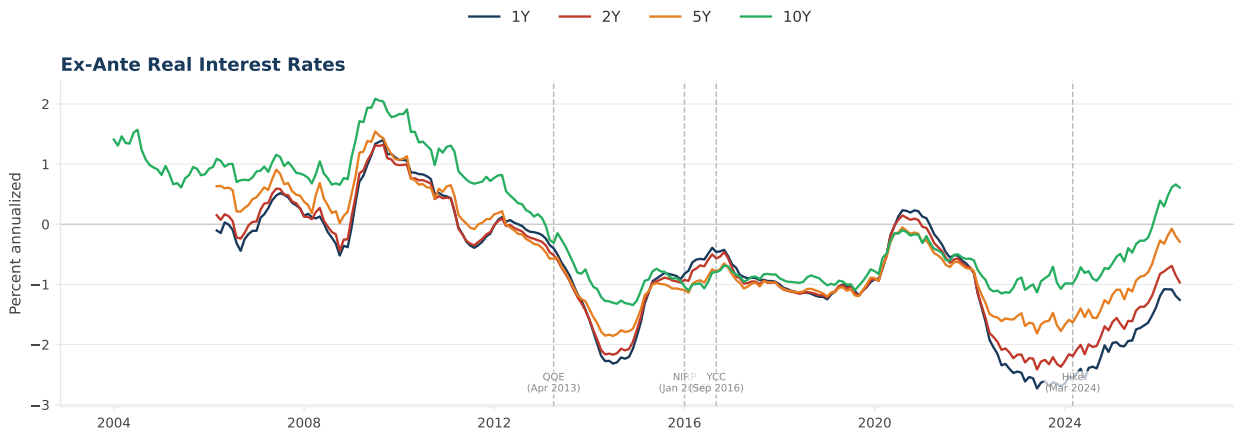
[20] The composite incorporates BEI as one of several inputs, so the resulting ex-ante real rate is not strictly survey-only. Survey sources nonetheless dominate the composite at horizons below five years, where BEI is unavailable, and contribute alongside BEI at the 5- and 10-year horizons.



(a) Model-implied ex-ante real rates and 95-percent confidence intervals, versus JGBi real yields at the 5-year and 10-year horizons.



(b) NSICX–JGBi real rate wedge and 95-percent confidence intervals at the 5-year and 10-year horizons, with the Christensen-Spiegel (CS) decomposition into inflation risk premium (IRP), liquidity premium (LP), and deflation protection (DP).



(c) Ex-ante real interest rates at the 1-year, 2-year, 5-year, and 10-year horizons, computed as nominal JGB yields minus NSICX inflation expectations.

Figure 9: Ex-ante real rates, JGBi real yields, and real-rate wedges.

percent. Following QQE in April 2013, real rates across all horizons declined and turned negative, with the real rate term structure compressing and eventually inverting as short-horizon rates fell more sharply than long-horizon rates. The short end reached approximately -2.7 percent during the post-pandemic episode as inflation expectations rose faster than nominal yields, while the 10-year remained between -0.5 and -1.2 percent. The March 2024 rate hike was followed by only a partial recovery in real rates, with the inversion persisting through mid-2026.

Additional decompositions of nominal yields into real rates and NSICX-implied inflation expectations around key BOJ policy events and across monetary policy regimes are reported in Appendix A, while the responses to high frequency-identified (HFI) BOJ monetary policy shocks are reported in Appendix B. The event study around key BOJ policy events indicates that QQE and NIRP announcements were associated with movements in inflation expectations across the term structure. On the other hand, the analysis using HFI shocks finds significant responses of inflation expectations mainly in the pre-QQE period, with limited evidence of a systematic response during the QQE/YCC era. This in turn suggests that, over the past decade, monetary policy transmitted primarily through the real-rate component of nominal yields, a necessary condition for the effective transmission to the real economy documented in the literature (Carrière-Swallow and others, 2025; Nakashima and others, 2024).

5 Conclusion

This paper develops the NSICX model, an internally consistent dynamic Nelson-Siegel framework for estimating the term structures of inflation expectations from multiple survey and market-based sources. The transition dynamics are derived to be uniquely consistent with the Nelson-Siegel loading structure, and measurement error variances are calibrated to historical forecast performance, reducing estimation to four free parameters. The model delivers coherent real-time estimates of the term structure across all horizons up to ten years. Applied to Japan, our model documents a gradual alignment of inflation expectations toward the 2-percent target from below since 2022, a strengthening of long-run expectation anchoring in sensitivities, and ex-ante real rates that turned negative around the time of the introduction of quantitative and qualitative easing.

Several extensions remain for future work. One direction is to integrate the NSICX composite with arbitrage-free models of nominal and real yields. Standard no-arbitrage frameworks identify term premia as the residual between observed yields and risk-neutral expectations, but this decomposition is sensitive to the assumed dynamics of the pricing factors. The persistent and time-varying behavior of inflation expectations documented here suggests that stationarity assumptions may distort term premia estimates over long samples, and that disciplining the physical measure with survey-based composites could yield a more robust decomposition of nominal yields into expectations, inflation risk premia, and real term premia.

More broadly, the internal consistency requirement derived here is relevant to any Nelson-Siegel term structure of physical-measure expectations, and the NSICX framework can be applied wherever survey data span multiple horizons of the same underlying variable, suggesting natural

extensions to expectations of interest rates, growth rates, and other macroeconomic variables. In future work, we also intend to extend the framework to a broad panel of advanced and emerging economies, where its flexibility in accommodating heterogeneous, irregularly reported data would support systematic cross-country comparison of the anchoring of inflation expectations.

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A Fisher decomposition around BoJ policy events

This appendix decomposes changes in nominal JGB yields into changes in real rates and changes in NSICX-implied inflation expectations around BoJ monetary policy events. Taking first differences of (16) in Section 4.5, the Fisher decomposition is

$$\Delta i_t(h) = \Delta r_t(h) + \Delta \hat{\pi}^e_t(h), \tag{17}$$

where changes are measured from the month prior to the announcement $(t-1)$ to the month following $(t+1)$. We first report an event-study decomposition around four key BoJ events (Appendix A.1), then a daily- and weekly-frequency robustness check on the inflation expectation component (Appendix A.2).

A.1 Event-study decomposition around key BoJ events

Figure 10 applies the decomposition in (17) to four key BoJ policy events: the introduction of QQE (April 2013), NIRP (January 2016), YCC (September 2016), and the March 2024 rate hike. For each event, we report the contribution of real rates and inflation expectations to changes in nominal JGB yields at the 1-, 2-, 5-, and 10-year horizons.

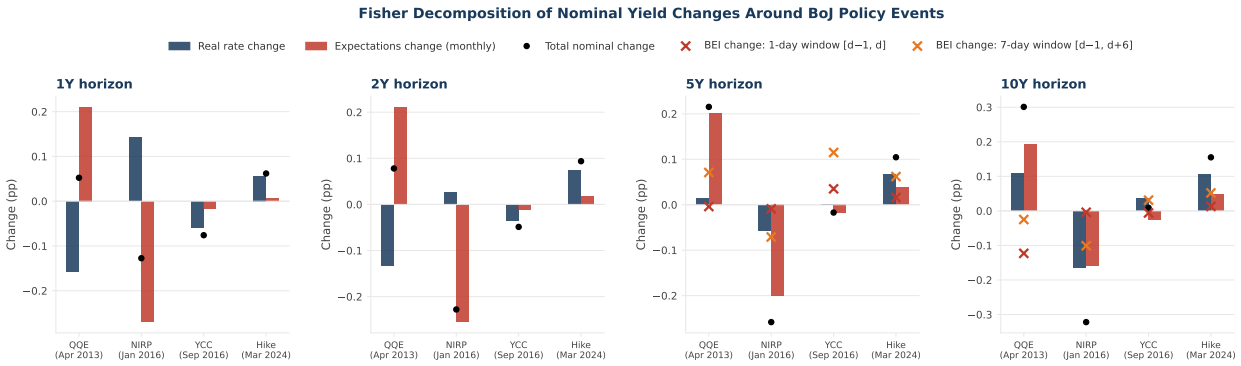


Figure 10: Fisher decomposition of JGB yield changes around key BoJ policy events at the 1-, 2-, 5-, and 10-year horizons. Blue bars show real rate contributions, red bars show estimated inflation expectations contributions, and black dots show total nominal changes.

For QQE, the decomposition highlights a positive expectations contribution of about 0.2 percentage points across horizons, while nominal yields were broadly unchanged at short horizons and increased at longer horizons, resulting in lower short-term real rates but higher long-term real rates in the announcement window. The subsequent multi-year decline in long-term real rates therefore reflects forces that unfolded beyond the immediate window, including the gradual compression of term premia as the BoJ's balance sheet expanded. For NIRP, the negative adjustment in nominal yields was partly offset by a decline in inflation expectations, with real rates increasing at short horizons and decreasing at longer horizons. YCC effects appear muted and mixed across horizons for both nominal and real yields. For the March 2024 hike, both nominal and real rates increased

in tandem, with little change in inflation expectations across horizons.

A.2 Daily and weekly BEI robustness

The event-study decomposition above is constructed over a two-month window, which risks conflating the impact of policy announcements with concurrent macroeconomic developments. This choice was necessitated by the nature of our data, since nominal yields are monthly averages to match the timing of the surveys, which are likewise reported throughout the calendar month and therefore reflect average expectations within a given month. The announcement month consequently blends pre- and post-announcement trading days and cannot serve as a clean baseline or endpoint. To assess robustness, we complement the monthly decomposition with daily and weekly changes in breakeven inflation at the 5- and 10-year horizons, computed over 1-day and 7-day windows starting from the day before the announcement; these estimates are reported alongside the monthly decomposition in Figure 10.

The BEI-based identification rests on three assumptions: that the window is tight enough to exclude other drivers of inflation expectations, that it is wide enough to capture the full response to the announcement, and that inflation risk and liquidity premia embedded in BEI are approximately stable over the window, so that BEI movements primarily reflect shifts in expectations. The window length therefore involves a tradeoff. A shorter window better excludes other news, while a longer window gives markets more time to react. Accordingly, we report estimates across multiple event windows, from 30 minutes (Appendix B), to 1 day, 7 days, and our baseline monthly decomposition, to assess the sensitivity of the results to alternative identification windows (e.g., Gürkaynak and others, 2005; Emeksiz and others, 2026). Finally, the third assumption, that premia are stable, is the more demanding one in the Japanese context, where Section 4.5 documents that the NSICX–JGBi wedge shifts materially around QQE (at the 10-year horizon) and NIRP (at both horizons), and to a lesser extent around YCC at the 5-year horizon. The daily and weekly estimates should therefore be interpreted with particular caution for these episodes. Excluding them, changes over the 7-day window are broadly consistent with the monthly inflation expectation changes from our composite term structure.

B Pass-through of BOJ policy shocks to inflation expectations

B.1 Specification

Let $y_t(\tau)$ denote the NSICX-implied average expected inflation rate over $[0, \tau]$ months, computed from the filtered Nelson–Siegel factors (Section 2.4) using the standard Nelson–Siegel forward loadings. We estimate Jordà (2005) cumulative-response local projections of the form

$$y_{t+h}(\tau) - y_{t-1}(\tau) \;=\; \alpha_h + \beta_h^{\mathrm{FG}}\,\mathrm{FG}_t + \beta_h^{\mathrm{LSAP}}\,\mathrm{LSAP}_t + \theta_h\,\mathrm{short}_t + u_{t+h}, \qquad h = 0, 1, \ldots, 24, \quad (18)$$

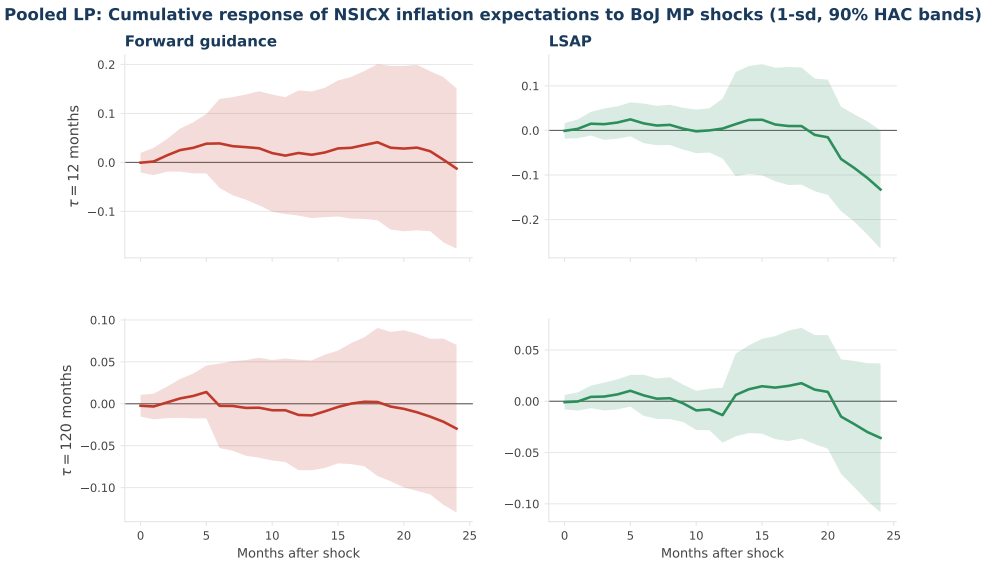
for the two focal maturities $\tau \in \{12, 120\}$ months. The explanatory variables are the orthogonalised 30-minute BOJ monetary policy surprises identified at MPM dates documented in Carrière-Swallow and others (2025): a forward-guidance factor (FG_t), a large-scale asset-purchase factor (LSAP_t), and the residual variation in surprises affecting short maturities (short_t), each pre-normalised to unit variance at the event level and aggregated to monthly frequency by summing within-month surprises. The short-rate factor is included as a control to absorb its non-trivial monthly correlation with FG and LSAP (e.g., $\rho(\mathrm{FG}_t, \mathrm{LSAP}_t) \approx -0.6$), but its coefficient is not reported since this factor does not have any meaningful economic interpretation. The estimation sample runs from January 2010 to December 2023 (168 monthly observations).

B.2 Estimation Results

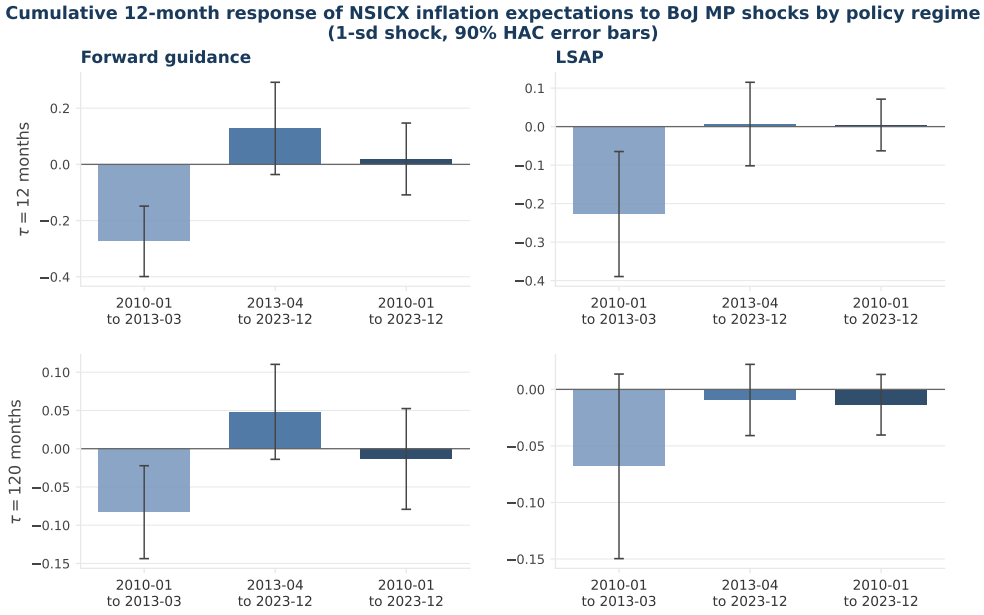
Figure 11a reports the pooled-sample cumulative responses of NSICX inflation expectations to FG and LSAP shocks at $\tau = 12$ and $\tau = 120$ months. The responses are statistically indistinguishable from zero at the 90% level through 12 months. At the 24-month horizon, an LSAP shock lowers 12-month expectations by approximately 13 basis points ($p = 0.10$), whereas the other responses remain statistically insignificant. The $\tau = 120$ responses are smaller in magnitude than the $\tau = 12$ responses.

The limited responsiveness of inflation expectations to monetary policy shocks, however, may partly reflect averaging across qualitatively different policy regimes. Figure 11b reports the $h = 12$ cumulative responses re-estimated on two non-overlapping subsamples corresponding to the BOJ's principal post-2010 frameworks: a pre-QQE window (January 2010 to March 2013, 39 months) covering the ZLB regime prior to the introduction of QQE, and a QQE/YCC window (April 2013 to December 2023, 129 months) spanning QQE, NIRP, and YCC.

In the pre-QQE regime, both hawkish forward-guidance and LSAP surprises lower NSICX expectations across maturities. A one-standard-deviation FG shock lowers expectations by approximately 27 basis points at $\tau = 12$ and 8 basis points at $\tau = 120$, while the corresponding LSAP responses are approximately 23 and 7 basis points. The signs are consistent with the standard monetary policy channel, where an upward revision of the expected policy-rate path at the ZLB depresses expected real activity and hence expected inflation. By contrast, in the consolidated QQE/YCC regime, the FG and LSAP responses are not statistically significant at either maturity.



(a) Pooled cumulative response, $h = 0, \ldots, 24$



(b) Cumulative 12-month response by policy regime

Figure 11: Cumulative response of NSICX inflation expectations to BOJ monetary policy shocks, estimated from Equation (18). Top row: $\tau = 12$ months; bottom row: $\tau = 120$ months. Left: forward-guidance shock; right: LSAP shock. Units are percentage points per one-standard-deviation shock, with 90% HAC bands. *Panel (a):* pooled sample, January 2010 to December 2023. *Panel (b):* $\beta^{h=12}$ by subsample: pre-QQE (January 2010 to March 2013, 39 months), QQE/YCC (April 2013 to December 2023, 129 months), and the full estimation sample.



PUBLICATIONS

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